



EDUCORE CAMPUS OS

# Roles.pdf

Seven-role engineering ownership model with exhaustive file-by-file coverage

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**Evidence base: complete project repository scan**  
**Edition: 188-file ownership register - detailed edition**  
**Prepared: 02 July 2026**

This report describes the implemented system as found in the repository.  
Recommendations and commercial figures are explicitly marked as planning proposals.

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# 1. Ownership objective and inventory rules

This document divides the complete EduCore project into exactly seven engineering roles: two frontend roles and five backend/database roles. Every project-owned file in the inventory has one primary owner so that work cannot disappear between teams. Primary ownership does not mean unilateral authority: cross-domain changes still require review from affected roles.

|                   |                             |                             |                                  |
|-------------------|-----------------------------|-----------------------------|----------------------------------|
| 7                 | 2 + 5                       | 188                         | 0                                |
| Engineering roles | Frontend + backend/database | Files assigned exactly once | Unowned or duplicate-owned files |

**Inventory boundary: includes all 184 version-controlled files plus five meaningful project files present in the workspace: CURRICULUM\_README.md, DEPLOYMENT.md, PPT.md, SIDEBAR\_ROLES\_ANALYSIS.md, and backend/alembic/versions/ \_\_init\_\_.py. It excludes .git internals, node\_modules, the Python virtual environment, caches, compiled/build output, generated PDF output, ignored runtime logs, and backend/.env because those are dependencies, transient artifacts, or local secrets.**

Two historical log rotations and seed\_output.txt are version-controlled and are therefore included under Role 7, with an explicit recommendation to remove generated operational output from source control.

## 2. Seven-role organization

| # | Role                                              | Mission                                                                                                                                                                      | Files |
|---|---------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 1 | Frontend Platform and Design System Engineer      | Own the browser application shell, authentication state, reusable UI primitives, global styling, build quality, and frontend delivery.                                       | 26    |
| 2 | Frontend Product Workflows Engineer               | Own feature pages, role-specific workspace behavior, resource metadata, forms, and the user journeys for academic, learning, finance, and operations.                        | 17    |
| 3 | Backend API, Identity and Security Engineer       | Own application assembly, authentication, authorization, request contracts, exception behavior, audit controls, dashboards, notifications, and cross-domain API integration. | 34    |
| 4 | Academic Domain and People Data Engineer          | Own the canonical academic hierarchy, curriculum rules, student/faculty identity, subject catalogue, HOD scope, and academic database evolution.                             | 26    |
| 5 | Learning, Assessment and Timetable Engineer       | Own attendance, marks, assignments, submissions, faculty teaching context, timetable versions, scheduling, and academic reports.                                             | 21    |
| 6 | Finance and Library Backend Engineer              | Own fee/invoice/payment/expense/salary behavior and the library catalogue, circulation, reservations, and fines.                                                             | 10    |
| 7 | Database Reliability, DevOps and Quality Engineer | Own database foundations, repositories, seeding and maintenance, migration operations, deployment assets, tests, logs, and technical documentation.                          | 54    |

### Why this split works

- The two frontend roles separate platform consistency from domain workflow delivery, avoiding a single UI bottleneck.
- Identity/security is separated from academic rules because access policy and curriculum policy fail in different ways.
- Academic people/data is separated from daily learning and timetables so foundational hierarchy changes receive dedicated review.
- Finance and library share one backend role because both are operational bounded contexts with transactional state and specialized permissions.

- Database reliability/DevOps/QA owns the lifecycle around all domains: migrations, seed data, repositories, deployment, verification, and documentation.
- Files that span multiple domains, especially `api_routes.py` and `seed.py`, have one accountable owner plus mandatory reviewers.

### 3. Decision rights and collaboration contracts

| Change type                           | Primary owner                 | Mandatory reviewers                                                       |
|---------------------------------------|-------------------------------|---------------------------------------------------------------------------|
| New page or workflow                  | Role 2                        | Role 1 for shared UI; relevant backend domain role                        |
| Shared UI/API client/auth state       | Role 1                        | Role 3 for session/security changes                                       |
| New endpoint or permission            | Role 3 or domain router owner | Role 3 plus consuming frontend role                                       |
| Academic schema/rule                  | Role 4                        | Role 7 for migration; Role 2 for forms/pages; Role 5 if teaching affected |
| Attendance/marks/assignment/timetable | Role 5                        | Role 4 for academic scope; Role 2 for UX; Role 7 for migration            |
| Money or library circulation state    | Role 6                        | Role 3 for permission/audit; Role 7 for migration/tests                   |
| Migration, seed, deployment, backup   | Role 7                        | Every domain owner whose data is touched                                  |

**Rule: A primary owner may edit every assigned file, but may not change a public API, permission boundary, persisted data meaning, financial state transition, or production operation without the listed reviewers.**

### 4. Feature ownership matrix

| Feature                                                       | Lead   | Supporting roles                  |
|---------------------------------------------------------------|--------|-----------------------------------|
| Authentication, RBAC, CAPTCHA, audit                          | Role 3 | Role 1, Role 7                    |
| Responsive shell and design system                            | Role 1 | Role 2                            |
| Generic resource screens and forms                            | Role 2 | Role 1, domain backend owner      |
| Departments, courses, branches, curriculum, students, faculty | Role 4 | Role 2, Role 3, Role 7            |
| Attendance, marks, assignments, timetable                     | Role 5 | Role 2, Role 4, Role 7            |
| Fees, collections, payments, expenses, salaries               | Role 6 | Role 2, Role 3, Role 7            |
| Library catalogue and circulation                             | Role 6 | Role 2, Role 3, Role 7            |
| Database lifecycle, seed, backup, migration, deployment       | Role 7 | All domain owners                 |
| Dashboards and notifications                                  | Role 3 | Role 2 and all data-domain owners |

## 5. Role 1 - Frontend Platform and Design System Engineer

Own the browser application shell, authentication state, reusable UI primitives, global styling, build quality, and frontend delivery.

| Accountability | Definition                                                                                                        |
|----------------|-------------------------------------------------------------------------------------------------------------------|
| Owens          | Implementation quality, tests/checks, documentation, review routing, and operational readiness for assigned files |
| May change     | The behavior described in each file card, within API/data/security review rules                                   |
| Must protect   | Role boundaries, backward compatibility, accessibility, data integrity, and traceability                          |
| File count     | 26                                                                                                                |

### Role feature responsibilities

- Application boot, protected navigation, session state, shared requests, reusable components, global visual language, lint/build, and frontend container delivery.
- Frontend performance, accessibility, responsive behavior, and consistent empty/loading/error states.

### Every file owned by this role

#### 146. frontend/.dockerignore

**Purpose:** Excludes local dependencies, build output, logs, and editor artifacts from the frontend container build context.

**Feature contribution:** Supports a consistent, secure, responsive browser application and repeatable frontend build/deployment.

**What this role can edit/change:** Can change the support behavior described above; coordinate with dependent files and run the relevant build/test workflow.

**Change caution:** Review downstream imports and the role boundary before merging.

#### 147. frontend/Dockerfile

**Purpose:** Builds the React/Vite application in a Node stage and serves the production assets from an Nginx stage.

**Feature contribution:** Supports a consistent, secure, responsive browser application and repeatable frontend build/deployment.

**What this role can edit/change:** Can change build, runtime, networking, proxying, migration, or quality behavior. Validate both local and production-like environments.

**Change caution:** Review downstream imports and the role boundary before merging.

#### 148. frontend/eslint.config.js

**Purpose:** Configures JavaScript and React hook/refresh lint rules for frontend code quality.

**Feature contribution:** Supports a consistent, secure, responsive browser application and repeatable frontend build/deployment.

**What this role can edit/change:** Can change build, runtime, networking, proxying, migration, or quality behavior. Validate both local and production-like environments.

**Change caution:** Review downstream imports and the role boundary before merging.

#### 149. frontend/index.html

**Purpose:** Browser entry document containing the application mount point and metadata used by Vite.

**Feature contribution:** Supports a consistent, secure, responsive browser application and repeatable frontend build/deployment.

**What this role can edit/change:** Can change the support behavior described above; coordinate with dependent files and run the relevant build/test workflow.

**Change caution:** Review downstream imports and the role boundary before merging.

**150. frontend/nginx.conf**

**Purpose:** Configures Nginx static serving, single-page application route fallback, and backend API/auth proxying.

**Feature contribution:** Supports a consistent, secure, responsive browser application and repeatable frontend build/deployment.

**What this role can edit/change:** Can change build, runtime, networking, proxying, migration, or quality behavior. Validate both local and production-like environments.

**Change caution:** Review downstream imports and the role boundary before merging.

**151. frontend/package-lock.json**

**Purpose:** Pins the complete frontend dependency graph for reproducible installation and builds.

**Feature contribution:** Supports a consistent, secure, responsive browser application and repeatable frontend build/deployment.

**What this role can edit/change:** Can change dependencies, scripts, and reproducible versions. Rebuild, lint, test, and review security advisories after edits.

**Change caution:** Generated lockfile; update through the package manager, not manual editing.

**152. frontend/package.json**

**Purpose:** Declares frontend scripts and runtime/development dependencies for React, routing, charts, icons, PDF/CSV support, Vite, and linting.

**Feature contribution:** Supports a consistent, secure, responsive browser application and repeatable frontend build/deployment.

**What this role can edit/change:** Can change dependencies, scripts, and reproducible versions. Rebuild, lint, test, and review security advisories after edits.

**Change caution:** Review downstream imports and the role boundary before merging.

**153. frontend/public/institute.png**

**Purpose:** EduCore institute/brand image used by the frontend and project documentation.

**Feature contribution:** Supports a consistent, secure, responsive browser application and repeatable frontend build/deployment.

**What this role can edit/change:** Can change the support behavior described above; coordinate with dependent files and run the relevant build/test workflow.

**Change caution:** Preserve aspect ratio, licensing provenance, and optimized file size.

**154. frontend/src/App.jsx**

**Purpose:** Defines protected client routes, lazy feature loading, HOD filtering, permission gates, and fallback navigation.

**Feature contribution:** Supports a consistent, secure, responsive browser application and repeatable frontend build/deployment. Code evidence: components/exports: ProtectedApp, App; routes: /, /dashboard, /hod-panel, /hod-management, `/\${config.slug}/assignments, /profile, /timetable, /notifications, \*, /login, /\*.

**What this role can edit/change:** Can change the support behavior described above; coordinate with dependent files and run the relevant build/test workflow.

**Change caution:** Review downstream imports and the role boundary before merging.

**155. frontend/src/components/layout/AppShell.jsx**

**Purpose:** Implements responsive navigation, role-filtered resources, workspace search, notifications, account menu, and sign-out.

**Feature contribution:** Supports a consistent, secure, responsive browser application and repeatable frontend build/deployment. Code evidence: components/exports: AppShell.

**What this role can edit/change:** Can change shared interaction and presentation across all consuming pages. Test keyboard, error, empty, and responsive states.

**Change caution:** Review downstream imports and the role boundary before merging.

**156. frontend/src/components/layout/PermissionGate.jsx**

**Purpose:** Project support file for PermissionGate.

**Feature contribution:** Supports a consistent, secure, responsive browser application and repeatable frontend build/deployment. Code evidence: components/exports: PermissionGate.

**What this role can edit/change:** Can change shared interaction and presentation across all consuming pages. Test keyboard, error, empty, and responsive states.

**Change caution:** Frontend visibility must never replace backend enforcement.

**157. frontend/src/components/ui/Button.jsx**

**Purpose:** Reusable interface component for Button, shared across multiple pages.

**Feature contribution:** Supports a consistent, secure, responsive browser application and repeatable frontend build/deployment. Code evidence: components/exports: Button.

**What this role can edit/change:** Can change shared interaction and presentation across all consuming pages. Test keyboard, error, empty, and responsive states.

**Change caution:** Review downstream imports and the role boundary before merging.

**158. frontend/src/components/ui/Card.jsx**

**Purpose:** Reusable interface component for Card, shared across multiple pages.

**Feature contribution:** Supports a consistent, secure, responsive browser application and repeatable frontend build/deployment. Code evidence: components/exports: Card, CardHeader, CardTitle, CardContent.

**What this role can edit/change:** Can change shared interaction and presentation across all consuming pages. Test keyboard, error, empty, and responsive states.

**Change caution:** Review downstream imports and the role boundary before merging.

**159. frontend/src/components/ui/ConfirmDialog.jsx**

**Purpose:** Reusable interface component for ConfirmDialog, shared across multiple pages.

**Feature contribution:** Supports a consistent, secure, responsive browser application and repeatable frontend build/deployment. Code evidence: components/exports: ConfirmDialog.

**What this role can edit/change:** Can change shared interaction and presentation across all consuming pages. Test keyboard, error, empty, and responsive states.

**Change caution:** Review downstream imports and the role boundary before merging.

**160. frontend/src/components/ui/PageHeader.jsx**

**Purpose:** Reusable interface component for PageHeader, shared across multiple pages.

**Feature contribution:** Supports a consistent, secure, responsive browser application and repeatable frontend build/deployment. Code evidence: components/exports: PageHeader.

**What this role can edit/change:** Can change shared interaction and presentation across all consuming pages. Test keyboard, error, empty, and responsive states.

**Change caution:** Review downstream imports and the role boundary before merging.

**161. frontend/src/components/ui/PeopleDirectory.jsx**

**Purpose:** Reusable interface component for PeopleDirectory, shared across multiple pages.

**Feature contribution:** Supports a consistent, secure, responsive browser application and repeatable frontend build/deployment. Code evidence: components/exports: PeopleDirectory.

**What this role can edit/change:** Can change shared interaction and presentation across all consuming pages. Test keyboard, error, empty, and responsive states.

**Change caution:** Review downstream imports and the role boundary before merging.

**162. frontend/src/components/ui/ProfileDetailModal.jsx**

**Purpose:** Reusable interface component for ProfileDetailModal, shared across multiple pages.

**Feature contribution:** Supports a consistent, secure, responsive browser application and repeatable frontend build/deployment. Code evidence: components/exports: ProfileDetailModal.

**What this role can edit/change:** Can change shared interaction and presentation across all consuming pages. Test keyboard, error, empty, and responsive states.

**Change caution:** Review downstream imports and the role boundary before merging.

**163. frontend/src/components/ui/RecordFormModal.jsx**

**Purpose:** Reusable interface component for RecordFormModal, shared across multiple pages.

**Feature contribution:** Supports a consistent, secure, responsive browser application and repeatable frontend build/deployment. Code evidence: components/exports: RecordFormModal, renderField, capitalize.

**What this role can edit/change:** Can change shared interaction and presentation across all consuming pages. Test keyboard, error, empty, and responsive states.

**Change caution:** Review downstream imports and the role boundary before merging.



**164. frontend/src/components/ui/StatCard.jsx**

**Purpose:** Reusable interface component for StatCard, shared across multiple pages.

**Feature contribution:** Supports a consistent, secure, responsive browser application and repeatable frontend build/deployment. Code evidence: components/exports: StatCard.

**What this role can edit/change:** Can change shared interaction and presentation across all consuming pages. Test keyboard, error, empty, and responsive states.

**Change caution:** Review downstream imports and the role boundary before merging.

**168. frontend/src/lib/api.js**

**Purpose:** Credentialed JSON request wrapper with standardized error extraction and GET/POST/PUT/DELETE helpers.

**Feature contribution:** Supports a consistent, secure, responsive browser application and repeatable frontend build/deployment. Code evidence: components/exports: request, api.

**What this role can edit/change:** Can change the support behavior described above; coordinate with dependent files and run the relevant build/test workflow.

**Change caution:** Review downstream imports and the role boundary before merging.

**169. frontend/src/main.jsx**

**Purpose:** Bootstraps React into the browser DOM and installs Router and AuthProvider around the application.

**Feature contribution:** Supports a consistent, secure, responsive browser application and repeatable frontend build/deployment.

**What this role can edit/change:** Can change the support behavior described above; coordinate with dependent files and run the relevant build/test workflow.

**Change caution:** Review downstream imports and the role boundary before merging.

**184. frontend/src/state/AuthContext.jsx**

**Purpose:** Restores the session, exposes login/logout, caches the current user, and provides frontend permission checks.

**Feature contribution:** Supports a consistent, secure, responsive browser application and repeatable frontend build/deployment. Code evidence: components/exports: AuthProvider, useAuth.

**What this role can edit/change:** Can change the support behavior described above; coordinate with dependent files and run the relevant build/test workflow.

**Change caution:** Review downstream imports and the role boundary before merging.

**185. frontend/src/styles/dashboard-widgets.css**

**Purpose:** Defines reusable visual rules and responsive styling for dashboard widgets.

**Feature contribution:** Supports a consistent, secure, responsive browser application and repeatable frontend build/deployment.

**What this role can edit/change:** Can change layout, color, typography, spacing, responsiveness, and visual states; check all breakpoints and shared selectors.

**Change caution:** Review downstream imports and the role boundary before merging.

**186. frontend/src/styles/index.css**

**Purpose:** Defines reusable visual rules and responsive styling for index.

**Feature contribution:** Supports a consistent, secure, responsive browser application and repeatable frontend build/deployment.

**What this role can edit/change:** Can change layout, color, typography, spacing, responsiveness, and visual states; check all breakpoints and shared selectors.

**Change caution:** Review downstream imports and the role boundary before merging.

**187. frontend/src/styles/people-directory.css**

**Purpose:** Defines reusable visual rules and responsive styling for people directory.

**Feature contribution:** Supports a consistent, secure, responsive browser application and repeatable frontend build/deployment.

**What this role can edit/change:** Can change layout, color, typography, spacing, responsiveness, and visual states; check all breakpoints and shared selectors.

**Change caution:** Review downstream imports and the role boundary before merging.

**188. frontend/vite.config.js**

**Purpose:** Configures the Vite React plugin, development server, and backend proxy behavior.

**Feature contribution:** Supports a consistent, secure, responsive browser application and repeatable frontend build/deployment.

**What this role can edit/change:** Can change build, runtime, networking, proxying, migration, or quality behavior. Validate both local and production-like environments.

**Change caution:** Review downstream imports and the role boundary before merging.

## 6. Role 2 - Frontend Product Workflows Engineer

Own feature pages, role-specific workspace behavior, resource metadata, forms, and the user journeys for academic, learning, finance, and operations.

| Accountability | Definition                                                                                                        |
|----------------|-------------------------------------------------------------------------------------------------------------------|
| Owns           | Implementation quality, tests/checks, documentation, review routing, and operational readiness for assigned files |
| May change     | The behavior described in each file card, within API/data/security review rules                                   |
| Must protect   | Role boundaries, backward compatibility, accessibility, data integrity, and traceability                          |
| File count     | 17                                                                                                                |

### Role feature responsibilities

- All persona workflows and feature pages, declarative resource tables/forms, role workspace rules, exports, feature-specific presentation, and end-user usability.
- Translate backend capability into safe, discoverable workflows without duplicating security policy.

### Every file owned by this role

#### 165. frontend/src/config/resourceForms.js

**Purpose:** Declarative create/edit form definitions with validation, dependent selectors, permissions, and field behavior.

**Feature contribution:** Adds or configures a visible user workflow, role-specific navigation, form behavior, or feature presentation. Code evidence: components/exports: resourceForms.

**What this role can edit/change:** Can change navigation visibility, endpoints, table columns, permissions, labels, selectors, and form validation across many screens.

**Change caution:** Review downstream imports and the role boundary before merging.

#### 166. frontend/src/config/resources.jsx

**Purpose:** Central catalogue of frontend resource workspaces, endpoints, permissions, columns, labels, and icons.

**Feature contribution:** Adds or configures a visible user workflow, role-specific navigation, form behavior, or feature presentation. Code evidence: components/exports: resourceConfigs, getValue.

**What this role can edit/change:** Can change navigation visibility, endpoints, table columns, permissions, labels, selectors, and form validation across many screens.

**Change caution:** Review downstream imports and the role boundary before merging.

#### 167. frontend/src/config/roleAccess.js

**Purpose:** Applies student and teacher workspace allow-lists, role-specific labels, and assignment/timetable access checks.

**Feature contribution:** Adds or configures a visible user workflow, role-specific navigation, form behavior, or feature presentation. Code evidence: components/exports: isStudentWorkspace, isTeacherWorkspace, canAccessResourceForRole, resourceLabelForRole, canAccessAssignments, canAccessTimetable.

**What this role can edit/change:** Can change navigation visibility, endpoints, table columns, permissions, labels, selectors, and form validation across many screens.

**Change caution:** Frontend visibility must never replace backend enforcement.

#### 170. frontend/src/pages/AssignmentsPage.css

**Purpose:** Provides the visual layout and responsive states for assignment creation, submission, review, and role-specific coursework experience.

**Feature contribution:** Adds or configures a visible user workflow, role-specific navigation, form behavior, or feature presentation.

**What this role can edit/change:** Can change the visible workflow, queries, mutations, loading/error states, and role experience. Keep API contracts, accessibility, and responsive behavior aligned.

**Change caution:** Review downstream imports and the role boundary before merging.

**171. frontend/src/pages/AssignmentsPage.jsx**

**Purpose:** Implements assignment creation, submission, review, and role-specific coursework experience.

**Feature contribution:** Adds or configures a visible user workflow, role-specific navigation, form behavior, or feature presentation. Code evidence: components/exports: AssignmentsPage.

**What this role can edit/change:** Can change the visible workflow, queries, mutations, loading/error states, and role experience. Keep API contracts, accessibility, and responsive behavior aligned.

**Change caution:** Review downstream imports and the role boundary before merging.

**172. frontend/src/pages/AttendancePage.css**

**Purpose:** Provides the visual layout and responsive states for attendance filtering, entry, editing, and learning record presentation.

**Feature contribution:** Adds or configures a visible user workflow, role-specific navigation, form behavior, or feature presentation.

**What this role can edit/change:** Can change the visible workflow, queries, mutations, loading/error states, and role experience. Keep API contracts, accessibility, and responsive behavior aligned.

**Change caution:** Review downstream imports and the role boundary before merging.

**173. frontend/src/pages/AttendancePage.jsx**

**Purpose:** Implements attendance filtering, entry, editing, and learning record presentation.

**Feature contribution:** Adds or configures a visible user workflow, role-specific navigation, form behavior, or feature presentation. Code evidence: components/exports: StudentAttendanceCalendar, fetchData, AttendancePage.

**What this role can edit/change:** Can change the visible workflow, queries, mutations, loading/error states, and role experience. Keep API contracts, accessibility, and responsive behavior aligned.

**Change caution:** Review downstream imports and the role boundary before merging.

**174. frontend/src/pages/DashboardPage.jsx**

**Purpose:** Implements persona-aware dashboard metrics, charts, and summary cards.

**Feature contribution:** Adds or configures a visible user workflow, role-specific navigation, form behavior, or feature presentation. Code evidence: components/exports: DashboardPage, AdminPanels, TeacherPanels, StudentPanels, HodPanels, AccountantPanels, LibrarianPanels, getGreeting, formatDate, EnrollmentCard, EmptyChart.

**What this role can edit/change:** Can change the visible workflow, queries, mutations, loading/error states, and role experience. Keep API contracts, accessibility, and responsive behavior aligned.

**Change caution:** Review downstream imports and the role boundary before merging.

**175. frontend/src/pages/FeeCollectionPage.css**

**Purpose:** Provides the visual layout and responsive states for cashier-focused student search, invoice selection, payment collection, and receipt workflow.

**Feature contribution:** Adds or configures a visible user workflow, role-specific navigation, form behavior, or feature presentation.

**What this role can edit/change:** Can change the visible workflow, queries, mutations, loading/error states, and role experience. Keep API contracts, accessibility, and responsive behavior aligned.

**Change caution:** Money-changing paths require idempotency, reconciliation, audit, and strict authorization.

**176. frontend/src/pages/FeeCollectionPage.jsx**

**Purpose:** Implements cashier-focused student search, invoice selection, payment collection, and receipt workflow.

**Feature contribution:** Adds or configures a visible user workflow, role-specific navigation, form behavior, or feature presentation. Code evidence: components/exports: FeeCollectionPage.

**What this role can edit/change:** Can change the visible workflow, queries, mutations, loading/error states, and role experience. Keep API contracts, accessibility, and responsive behavior aligned.

**Change caution:** Money-changing paths require idempotency, reconciliation, audit, and strict authorization.

**177. frontend/src/pages/HodManagementPage.jsx**

**Purpose:** Implements HOD dashboard, branch scope, leadership assignment, and academic management.

**Feature contribution:** Adds or configures a visible user workflow, role-specific navigation, form behavior, or feature presentation. Code evidence: components/exports: HodManagementPage.

**What this role can edit/change:** Can change the visible workflow, queries, mutations, loading/error states, and role experience. Keep API contracts, accessibility, and responsive behavior aligned.

**Change caution:** Review downstream imports and the role boundary before merging.

**178. frontend/src/pages/HodPanelPage.jsx**

**Purpose:** Implements HOD dashboard, branch scope, leadership assignment, and academic management.

**Feature contribution:** Adds or configures a visible user workflow, role-specific navigation, form behavior, or feature presentation. Code evidence: components/exports: HodPanelPage.

**What this role can edit/change:** Can change the visible workflow, queries, mutations, loading/error states, and role experience. Keep API contracts, accessibility, and responsive behavior aligned.

**Change caution:** Review downstream imports and the role boundary before merging.

**179. frontend/src/pages/LoginPage.jsx**

**Purpose:** Implements CAPTCHA-assisted sign-in and authentication error feedback.

**Feature contribution:** Adds or configures a visible user workflow, role-specific navigation, form behavior, or feature presentation. Code evidence: components/exports: LoginPage.

**What this role can edit/change:** Can change the visible workflow, queries, mutations, loading/error states, and role experience. Keep API contracts, accessibility, and responsive behavior aligned.

**Change caution:** Review downstream imports and the role boundary before merging.

**180. frontend/src/pages/NotificationsPage.jsx**

**Purpose:** Implements notification listing and mark-all-read behavior.

**Feature contribution:** Adds or configures a visible user workflow, role-specific navigation, form behavior, or feature presentation. Code evidence: components/exports: NotificationsPage.

**What this role can edit/change:** Can change the visible workflow, queries, mutations, loading/error states, and role experience. Keep API contracts, accessibility, and responsive behavior aligned.

**Change caution:** Review downstream imports and the role boundary before merging.

**181. frontend/src/pages/ProfilePage.jsx**

**Purpose:** Implements personal profile and password/account information.

**Feature contribution:** Adds or configures a visible user workflow, role-specific navigation, form behavior, or feature presentation. Code evidence: components/exports: ProfileSection, ProfilePage.

**What this role can edit/change:** Can change the visible workflow, queries, mutations, loading/error states, and role experience. Keep API contracts, accessibility, and responsive behavior aligned.

**Change caution:** Review downstream imports and the role boundary before merging.

**182. frontend/src/pages/ResourcePage.jsx**

**Purpose:** Implements generic paginated/filterable data table, export, create, edit, detail, and delete behavior.

**Feature contribution:** Adds or configures a visible user workflow, role-specific navigation, form behavior, or feature presentation. Code evidence: components/exports: ResourcePage.

**What this role can edit/change:** Can change the visible workflow, queries, mutations, loading/error states, and role experience. Keep API contracts, accessibility, and responsive behavior aligned.

**Change caution:** Review downstream imports and the role boundary before merging.

**183. frontend/src/pages/TimetablePage.jsx**

**Purpose:** Implements interactive timetable viewing/editing, version submission, and approval experience.

**Feature contribution:** Adds or configures a visible user workflow, role-specific navigation, form behavior, or feature presentation. Code evidence: components/exports: TimetablePage.

**What this role can edit/change:** Can change the visible workflow, queries, mutations, loading/error states, and role experience. Keep API contracts, accessibility, and responsive behavior aligned.

**Change caution:** Review downstream imports and the role boundary before merging.

## 7. Role 3 - Backend API, Identity and Security Engineer

Own application assembly, authentication, authorization, request contracts, exception behavior, audit controls, dashboards, notifications, and cross-domain API integration.

| Accountability | Definition                                                                                                        |
|----------------|-------------------------------------------------------------------------------------------------------------------|
| Owens          | Implementation quality, tests/checks, documentation, review routing, and operational readiness for assigned files |
| May change     | The behavior described in each file card, within API/data/security review rules                                   |
| Must protect   | Role boundaries, backward compatibility, accessibility, data integrity, and traceability                          |
| File count     | 34                                                                                                                |

### Role feature responsibilities

- API assembly and cross-domain routing, authentication, authorization, identity contracts, exception handling, audit, email, notifications, dashboards, and maintenance controls.
- Ensure every request is authenticated/scoped correctly and every public response is intentional.

### Every file owned by this role

#### 029. backend/app/\_\_init\_\_.py

**Purpose:** Marks the directory as a Python package and may expose package-level imports.

**Feature contribution:** Adds backend transport, identity, security, platform governance, or personalized API behavior.

**What this role can edit/change:** Can change the support behavior described above; coordinate with dependent files and run the relevant build/test workflow.

**Change caution:** Review downstream imports and the role boundary before merging.

#### 030. backend/app/cli.py

**Purpose:** Project support file for cli.

**Feature contribution:** Adds backend transport, identity, security, platform governance, or personalized API behavior. Code evidence: top-level functions: initialize\_database, main.

**What this role can edit/change:** Can change the support behavior described above; coordinate with dependent files and run the relevant build/test workflow.

**Change caution:** Review downstream imports and the role boundary before merging.

#### 031. backend/app/core/\_\_init\_\_.py

**Purpose:** Provides cross-cutting backend platform behavior for init .

**Feature contribution:** Adds backend transport, identity, security, platform governance, or personalized API behavior.

**What this role can edit/change:** Can change the support behavior described above; coordinate with dependent files and run the relevant build/test workflow.

**Change caution:** Review downstream imports and the role boundary before merging.

#### 032. backend/app/core/config.py

**Purpose:** Provides cross-cutting backend platform behavior for config.

**Feature contribution:** Adds backend transport, identity, security, platform governance, or personalized API behavior. Code evidence: classes: Settings.

**What this role can edit/change:** Can change the support behavior described above; coordinate with dependent files and run the relevant build/test workflow.

**Change caution:** Review downstream imports and the role boundary before merging.

**033. backend/app/core/dependencies.py**

**Purpose:** Provides cross-cutting backend platform behavior for dependencies.

**Feature contribution:** Adds backend transport, identity, security, platform governance, or personalized API behavior. Code evidence: top-level functions: `get_db`, `get_current_user`, `get_current_active_user`, `get_optional_user`.

**What this role can edit/change:** Can change the support behavior described above; coordinate with dependent files and run the relevant build/test workflow.

**Change caution:** Review downstream imports and the role boundary before merging.

**034. backend/app/core/events.py**

**Purpose:** Provides cross-cutting backend platform behavior for events.

**Feature contribution:** Adds backend transport, identity, security, platform governance, or personalized API behavior. Code evidence: classes: `Event`, `FeePaidEvent`, `EventListener`, `EmailListener`, `AuditListener`, `EventDispatcher`.

**What this role can edit/change:** Can change the support behavior described above; coordinate with dependent files and run the relevant build/test workflow.

**Change caution:** Review downstream imports and the role boundary before merging.

**035. backend/app/core/exception\_handlers.py**

**Purpose:** Provides cross-cutting backend platform behavior for exception handlers.

**Feature contribution:** Adds backend transport, identity, security, platform governance, or personalized API behavior. Code evidence: top-level functions: `register_exception_handlers`.

**What this role can edit/change:** Can change the support behavior described above; coordinate with dependent files and run the relevant build/test workflow.

**Change caution:** Review downstream imports and the role boundary before merging.

**036. backend/app/core/exceptions.py**

**Purpose:** Provides cross-cutting backend platform behavior for exceptions.

**Feature contribution:** Adds backend transport, identity, security, platform governance, or personalized API behavior. Code evidence: classes: `CMSException`, `BadRequestException`, `UnauthorizedException`, `ForbiddenException`, `NotFoundException`, `ConflictException`, `ValidationException`, `ServiceUnavailableException`.

**What this role can edit/change:** Can change the support behavior described above; coordinate with dependent files and run the relevant build/test workflow.

**Change caution:** Review downstream imports and the role boundary before merging.

**037. backend/app/core/logging\_config.py**

**Purpose:** Provides cross-cutting backend platform behavior for logging config.

**Feature contribution:** Adds backend transport, identity, security, platform governance, or personalized API behavior. Code evidence: top-level functions: `setup_logging`, `get_logger`.

**What this role can edit/change:** May be replaced by fresh diagnostics, but should not contain secrets or personal data. Prefer removal from version control and managed log retention.

**Change caution:** Review downstream imports and the role boundary before merging.

**038. backend/app/core/permissions.py**

**Purpose:** Implements backend permission and role enforcement dependencies.

**Feature contribution:** Adds backend transport, identity, security, platform governance, or personalized API behavior. Code evidence: classes: `PermissionChecker`, `RoleChecker`.

**What this role can edit/change:** Can change the support behavior described above; coordinate with dependent files and run the relevant build/test workflow.

**Change caution:** Frontend visibility must never replace backend enforcement.

**040. backend/app/core/security.py**

**Purpose:** Implements password hashing, JWT signing/verification, and CAPTCHA state.

**Feature contribution:** Adds backend transport, identity, security, platform governance, or personalized API behavior. Code evidence: classes: `PasswordHasher`, `JWTHandler`, `CaptchaHandler`.

**What this role can edit/change:** Can change the support behavior described above; coordinate with dependent files and run the relevant build/test workflow.

**Change caution:** Review downstream imports and the role boundary before merging.



**045. backend/app/main.py**

**Purpose:** Constructs the FastAPI application, lifespan behavior, middleware, CORS, exception handling, and all router registrations.

**Feature contribution:** Adds backend transport, identity, security, platform governance, or personalized API behavior. Code evidence: top-level functions: lifespan, create\_app; 1 HTTP operations.

**What this role can edit/change:** Can change the support behavior described above; coordinate with dependent files and run the relevant build/test workflow.

**Change caution:** Review downstream imports and the role boundary before merging.

**068. backend/app/middleware/\_\_init\_\_.py**

**Purpose:** Marks the directory as a Python package and may expose package-level imports.

**Feature contribution:** Adds backend transport, identity, security, platform governance, or personalized API behavior.

**What this role can edit/change:** Can change the support behavior described above; coordinate with dependent files and run the relevant build/test workflow.

**Change caution:** Review downstream imports and the role boundary before merging.

**069. backend/app/middleware/audit\_middleware.py**

**Purpose:** Project support file for audit middleware.

**Feature contribution:** Adds backend transport, identity, security, platform governance, or personalized API behavior. Code evidence: classes: AuditMiddleware.

**What this role can edit/change:** Can change the support behavior described above; coordinate with dependent files and run the relevant build/test workflow.

**Change caution:** Review downstream imports and the role boundary before merging.

**073. backend/app/models/audit\_log.py**

**Purpose:** Defines SQLAlchemy persistence entities and relationships for the audit log domain.

**Feature contribution:** Adds backend transport, identity, security, platform governance, or personalized API behavior. Code evidence: tables: audit\_logs; classes: AuditLog.

**What this role can edit/change:** Can change stored fields, relationships, constraints, cascades, and lifecycle behavior. Coordinate schemas, migrations, services, seed data, frontend forms, and tests.

**Change caution:** Data-loss and compatibility risk is high; backup and migration rehearsal are mandatory.

**079. backend/app/models/notification.py**

**Purpose:** Defines SQLAlchemy persistence entities and relationships for the notification domain.

**Feature contribution:** Adds backend transport, identity, security, platform governance, or personalized API behavior. Code evidence: tables: notifications; classes: NotificationType, Notification.

**What this role can edit/change:** Can change stored fields, relationships, constraints, cascades, and lifecycle behavior. Coordinate schemas, migrations, services, seed data, frontend forms, and tests.

**Change caution:** Data-loss and compatibility risk is high; backup and migration rehearsal are mandatory.

**085. backend/app/models/user.py**

**Purpose:** Defines SQLAlchemy persistence entities and relationships for the user domain.

**Feature contribution:** Adds backend transport, identity, security, platform governance, or personalized API behavior. Code evidence: tables: users, roles, permissions, user\_roles, role\_permissions; classes: GenderEnum, User, Role, Permission, UserRole, RolePermission.

**What this role can edit/change:** Can change stored fields, relationships, constraints, cascades, and lifecycle behavior. Coordinate schemas, migrations, services, seed data, frontend forms, and tests.

**Change caution:** Data-loss and compatibility risk is high; backup and migration rehearsal are mandatory.

**089. backend/app/routers/\_\_init\_\_.py**

**Purpose:** Publishes FastAPI HTTP endpoints for the init domain and connects permissions to services/data access.

**Feature contribution:** Adds backend transport, identity, security, platform governance, or personalized API behavior.

**What this role can edit/change:** Can add/change endpoints, filters, status codes, permissions, serialization, and orchestration. Preserve backend authorization and update clients/tests.

**Change caution:** Review downstream imports and the role boundary before merging.



**091. backend/app/routers/api\_routes.py**

**Purpose:** Large cross-domain REST router implementing people, departments, courses, subjects, attendance, marks, library, and timetable operations.

**Feature contribution:** Adds backend transport, identity, security, platform governance, or personalized API behavior. Code evidence: classes: AssignHODRequest, TimetableSlotSchema, TimetableSaveRequest, TimetableStatusRequest; top-level functions: \_serialize, list\_users, get\_user, create\_user, update\_user, delete\_user, list\_students, get\_student, \_format\_student, create\_student, update\_student, delete\_student, list\_teachers, \_format\_teacher; 65 HTTP operations.

**What this role can edit/change:** Can add/change endpoints, filters, status codes, permissions, serialization, and orchestration. Preserve backend authorization and update clients/tests.

**Change caution:** Review downstream imports and the role boundary before merging.

**093. backend/app/routers/auth.py**

**Purpose:** Publishes FastAPI HTTP endpoints for the auth domain and connects permissions to services/data access.

**Feature contribution:** Adds backend transport, identity, security, platform governance, or personalized API behavior. Code evidence: top-level functions: login, logout, get\_current\_user\_info, get\_profile, change\_password; 5 HTTP operations.

**What this role can edit/change:** Can add/change endpoints, filters, status codes, permissions, serialization, and orchestration. Preserve backend authorization and update clients/tests.

**Change caution:** Review downstream imports and the role boundary before merging.

**094. backend/app/routers/captcha.py**

**Purpose:** Publishes FastAPI HTTP endpoints for the captcha domain and connects permissions to services/data access.

**Feature contribution:** Adds backend transport, identity, security, platform governance, or personalized API behavior. Code evidence: top-level functions: get\_captcha, get\_captcha\_svg, new\_captcha; 3 HTTP operations.

**What this role can edit/change:** Can add/change endpoints, filters, status codes, permissions, serialization, and orchestration. Preserve backend authorization and update clients/tests.

**Change caution:** Review downstream imports and the role boundary before merging.

**095. backend/app/routers/dashboard.py**

**Purpose:** Publishes FastAPI HTTP endpoints for the dashboard domain and connects permissions to services/data access.

**Feature contribution:** Adds backend transport, identity, security, platform governance, or personalized API behavior. Code evidence: top-level functions: dashboard; 1 HTTP operations.

**What this role can edit/change:** Can add/change endpoints, filters, status codes, permissions, serialization, and orchestration. Preserve backend authorization and update clients/tests.

**Change caution:** Review downstream imports and the role boundary before merging.

**097. backend/app/routers/notifications.py**

**Purpose:** Publishes FastAPI HTTP endpoints for the notifications domain and connects permissions to services/data access.

**Feature contribution:** Adds backend transport, identity, security, platform governance, or personalized API behavior. Code evidence: top-level functions: list\_notifications, mark\_all\_read; 2 HTTP operations.

**What this role can edit/change:** Can add/change endpoints, filters, status codes, permissions, serialization, and orchestration. Preserve backend authorization and update clients/tests.

**Change caution:** Review downstream imports and the role boundary before merging.

**102. backend/app/schemas/auth.py**

**Purpose:** Defines validated request and response contracts for the auth domain.

**Feature contribution:** Adds backend transport, identity, security, platform governance, or personalized API behavior. Code evidence: classes: LoginRequest, TokenResponse, PasswordChangeRequest.

**What this role can edit/change:** Can change accepted payloads, validation, defaults, and returned fields. Treat response changes as API compatibility changes.

**Change caution:** Review downstream imports and the role boundary before merging.

**103. backend/app/schemas/common.py**

**Purpose:** Defines validated request and response contracts for the common domain.

**Feature contribution:** Adds backend transport, identity, security, platform governance, or personalized API behavior. Code evidence: classes: PaginationParams, SortParams, FilterParams, PaginatedResponse, MessageResponse, ErrorResponse.

**What this role can edit/change:** Can change accepted payloads, validation, defaults, and returned fields. Treat response changes as API compatibility changes.

**Change caution:** Review downstream imports and the role boundary before merging.

**105. backend/app/schemas/dashboard.py**

**Purpose:** Defines validated request and response contracts for the dashboard domain.

**Feature contribution:** Adds backend transport, identity, security, platform governance, or personalized API behavior. Code evidence: classes: DashboardStat, AdminDashboardData, TeacherDashboardData, StudentDashboardData, AccountantDashboardData, HODDashboardData, LibrarianDashboardData.

**What this role can edit/change:** Can change accepted payloads, validation, defaults, and returned fields. Treat response changes as API compatibility changes.

**Change caution:** Review downstream imports and the role boundary before merging.

**113. backend/app/schemas/user.py**

**Purpose:** Defines validated request and response contracts for the user domain.

**Feature contribution:** Adds backend transport, identity, security, platform governance, or personalized API behavior. Code evidence: classes: UserBase, UserCreate, UserUpdate, UserResponse, UserListResponse, RoleCreate, RoleUpdate, RoleResponse, PermissionCreate, PermissionResponse.

**What this role can edit/change:** Can change accepted payloads, validation, defaults, and returned fields. Treat response changes as API compatibility changes.

**Change caution:** Review downstream imports and the role boundary before merging.

**117. backend/app/services/auth\_service.py**

**Purpose:** Encapsulates business logic and reusable operations for auth service.

**Feature contribution:** Adds backend transport, identity, security, platform governance, or personalized API behavior. Code evidence: classes: AuthService.

**What this role can edit/change:** Can change business rules, transactions, calculations, workflow state, and domain validation. Add regression tests for every policy change.

**Change caution:** Review downstream imports and the role boundary before merging.

**118. backend/app/services/base.py**

**Purpose:** Encapsulates business logic and reusable operations for base.

**Feature contribution:** Adds backend transport, identity, security, platform governance, or personalized API behavior. Code evidence: classes: IService.

**What this role can edit/change:** Can change business rules, transactions, calculations, workflow state, and domain validation. Add regression tests for every policy change.

**Change caution:** Review downstream imports and the role boundary before merging.

**120. backend/app/services/dashboard\_service.py**

**Purpose:** Implements role-specific dashboard aggregation through a factory pattern.

**Feature contribution:** Adds backend transport, identity, security, platform governance, or personalized API behavior. Code evidence: classes: BaseDashboard, AdminDashboard, TeacherDashboard, StudentDashboard, AccountantDashboard, HODDashboard, LibrarianDashboard, DashboardFactory.

**What this role can edit/change:** Can change business rules, transactions, calculations, workflow state, and domain validation. Add regression tests for every policy change.

**Change caution:** Review downstream imports and the role boundary before merging.

**121. backend/app/services/email\_service.py**

**Purpose:** Encapsulates business logic and reusable operations for email service.

**Feature contribution:** Adds backend transport, identity, security, platform governance, or personalized API behavior. Code evidence: top-level functions: get\_mail\_config, is\_email\_service\_configured, send\_student\_credentials, send\_teacher\_credentials.

**What this role can edit/change:** Can change business rules, transactions, calculations, workflow state, and domain validation. Add regression tests for every policy change.

**Change caution:** Review downstream imports and the role boundary before merging.

**123. backend/app/services/maintenance\_service.py**

**Purpose:** Encapsulates business logic and reusable operations for maintenance service.

**Feature contribution:** Adds backend transport, identity, security, platform governance, or personalized API behavior. Code evidence: classes: SystemMaintenanceService.

**What this role can edit/change:** Can change business rules, transactions, calculations, workflow state, and domain validation. Add regression tests for every policy change.

**Change caution:** Review downstream imports and the role boundary before merging.

**124. backend/app/templates/email/student\_welcome.html**

**Purpose:** HTML template or browser entry document for student welcome.

**Feature contribution:** Adds backend transport, identity, security, platform governance, or personalized API behavior.

**What this role can edit/change:** Can change the support behavior described above; coordinate with dependent files and run the relevant build/test workflow.

**Change caution:** Review downstream imports and the role boundary before merging.

**125. backend/app/templates/email/teacher\_welcome.html**

**Purpose:** HTML template or browser entry document for teacher welcome.

**Feature contribution:** Adds backend transport, identity, security, platform governance, or personalized API behavior.

**What this role can edit/change:** Can change the support behavior described above; coordinate with dependent files and run the relevant build/test workflow.

**Change caution:** Review downstream imports and the role boundary before merging.

## 8. Role 4 - Academic Domain and People Data Engineer

Own the canonical academic hierarchy, curriculum rules, student/faculty identity, subject catalogue, HOD scope, and academic database evolution.

| Accountability | Definition                                                                                                        |
|----------------|-------------------------------------------------------------------------------------------------------------------|
| Owens          | Implementation quality, tests/checks, documentation, review routing, and operational readiness for assigned files |
| May change     | The behavior described in each file card, within API/data/security review rules                                   |
| Must protect   | Role boundaries, backward compatibility, accessibility, data integrity, and traceability                          |
| File count     | 26                                                                                                                |

### Role feature responsibilities

- Canonical academic hierarchy, curriculum versioning, student/faculty/subject identity, HOD scope, automatic subject resolution, and academic migrations/data tools.
- Protect academic invariants and prevent cohort, branch, semester, or identifier corruption.

### Every file owned by this role

#### 017. backend/alembic/versions/0002\_academic\_structure.py

**Purpose:** Versioned database migration for 0002 academic structure; changes persisted schema and/or backfills data.

**Feature contribution:** Adds academic truth, people records, curriculum behavior, or institution hierarchy. Code evidence: top-level functions: `audit_columns`, `upgrade`, `_add_fk_column`, `downgrade`.

**What this role can edit/change:** Can change schema history, constraints, indexes, cascades, and backfills. Never rewrite an applied production revision; add a new revision and test upgrade/downgrade.

**Change caution:** Production databases may already depend on the exact revision graph.

#### 019. backend/alembic/versions/0004\_null\_safe\_academic\_scopes.py

**Purpose:** Versioned database migration for 0004 null safe academic scopes; changes persisted schema and/or backfills data.

**Feature contribution:** Adds academic truth, people records, curriculum behavior, or institution hierarchy. Code evidence: top-level functions: `_scope_column`, `_inspector`, `_has_column`, `_has_index`, `_has_check`, `_ensure_scope`, `_replace_unique`, `upgrade`, `downgrade`.

**What this role can edit/change:** Can change schema history, constraints, indexes, cascades, and backfills. Never rewrite an applied production revision; add a new revision and test upgrade/downgrade.

**Change caution:** Production databases may already depend on the exact revision graph.

#### 020. backend/alembic/versions/0005\_canonical\_hierarchy\_and\_student\_sequences.py

**Purpose:** Versioned database migration for 0005 canonical hierarchy and student sequences; changes persisted schema and/or backfills data.

**Feature contribution:** Adds academic truth, people records, curriculum behavior, or institution hierarchy. Code evidence: top-level functions: `upgrade`, `downgrade`.

**What this role can edit/change:** Can change schema history, constraints, indexes, cascades, and backfills. Never rewrite an applied production revision; add a new revision and test upgrade/downgrade.

**Change caution:** Production databases may already depend on the exact revision graph.

#### 023. backend/alembic/versions/812be78ed321\_add\_hod\_id\_to\_branch\_manual.py

**Purpose:** Versioned database migration for 812be78ed321 add hod id to branch manual; changes persisted schema and/or backfills data.

**Feature contribution:** Adds academic truth, people records, curriculum behavior, or institution hierarchy. Code evidence: top-level functions: `upgrade`, `downgrade`.

**What this role can edit/change:** Can change schema history, constraints, indexes, cascades, and backfills. Never rewrite an applied production revision; add a new revision and test upgrade/downgrade.

**Change caution:** Production databases may already depend on the exact revision graph.

**055. backend/app/management/commands/migrate\_faculty\_ids.py**

**Purpose:** Administrative command for migrate faculty ids; used for controlled data repair, migration, seeding, verification, or cleanup.

**Feature contribution:** Adds academic truth, people records, curriculum behavior, or institution hierarchy. Code evidence: top-level functions: migrate.

**What this role can edit/change:** Can change one-off operational data behavior. Run only against backups/staging first and make destructive intent explicit.

**Change caution:** Review downstream imports and the role boundary before merging.

**056. backend/app/management/commands/migrate\_student\_ids.py**

**Purpose:** Administrative command for migrate student ids; used for controlled data repair, migration, seeding, verification, or cleanup.

**Feature contribution:** Adds academic truth, people records, curriculum behavior, or institution hierarchy. Code evidence: top-level functions: get\_branch\_code, migrate.

**What this role can edit/change:** Can change one-off operational data behavior. Run only against backups/staging first and make destructive intent explicit.

**Change caution:** Review downstream imports and the role boundary before merging.

**060. backend/app/management/commands/remove\_hod\_perms.py**

**Purpose:** Administrative command for remove hod perms; used for controlled data repair, migration, seeding, verification, or cleanup.

**Feature contribution:** Adds academic truth, people records, curriculum behavior, or institution hierarchy. Code evidence: top-level functions: remove\_hod\_permissions.

**What this role can edit/change:** Can change one-off operational data behavior. Run only against backups/staging first and make destructive intent explicit.

**Change caution:** Potentially destructive or data-replacing operation; never run against valuable data without a verified backup.

**062. backend/app/management/commands/revert\_5\_real\_life\_departments.py**

**Purpose:** Administrative command for revert 5 real life departments; used for controlled data repair, migration, seeding, verification, or cleanup.

**Feature contribution:** Adds academic truth, people records, curriculum behavior, or institution hierarchy. Code evidence: top-level functions: main.

**What this role can edit/change:** Can change one-off operational data behavior. Run only against backups/staging first and make destructive intent explicit.

**Change caution:** Potentially destructive or data-replacing operation; never run against valuable data without a verified backup.

**063. backend/app/management/commands/seed\_5\_real\_life\_departments.py**

**Purpose:** Administrative command for seed 5 real life departments; used for controlled data repair, migration, seeding, verification, or cleanup.

**Feature contribution:** Adds academic truth, people records, curriculum behavior, or institution hierarchy. Code evidence: top-level functions: main.

**What this role can edit/change:** Can change one-off operational data behavior. Run only against backups/staging first and make destructive intent explicit.

**Change caution:** Potentially destructive or data-replacing operation; never run against valuable data without a verified backup.

**064. backend/app/management/commands/seed\_all\_subjects.py**

**Purpose:** Administrative command for seed all subjects; used for controlled data repair, migration, seeding, verification, or cleanup.

**Feature contribution:** Adds academic truth, people records, curriculum behavior, or institution hierarchy. Code evidence: top-level functions: get\_base\_topics, generate\_subjects, main.

**What this role can edit/change:** Can change one-off operational data behavior. Run only against backups/staging first and make destructive intent explicit.

**Change caution:** Potentially destructive or data-replacing operation; never run against valuable data without a verified backup.

**067. backend/app/management/commands/verify\_subjects.py**

**Purpose:** Administrative command for verify subjects; used for controlled data repair, migration, seeding, verification, or cleanup.

**Feature contribution:** Adds academic truth, people records, curriculum behavior, or institution hierarchy.

**What this role can edit/change:** Can change one-off operational data behavior. Run only against backups/staging first and make destructive intent explicit.

**Change caution:** Review downstream imports and the role boundary before merging.

**071. backend/app/models/academic.py**

**Purpose:** Implements branches, curricula, regulation versions, semesters, sections, elective groups, curriculum mappings, faculty allocations, assignments, and submissions.

**Feature contribution:** Adds academic truth, people records, curriculum behavior, or institution hierarchy. Code evidence: tables: branches, subject\_types, curricula, curriculum\_versions, academic\_semesters, sections, elective\_groups, curriculum\_subjects, student\_electives, faculty\_assignments, assignments, assignment\_submissions; classes: Branch, SubjectType, Curriculum, CurriculumVersion, AcademicSemester, Section, ElectiveGroup, CurriculumSubject, StudentElective, FacultyAssignment, Assignment, AssignmentSubmission; top-level functions: \_sync\_section\_scope, \_sync\_curriculum\_subject\_scope, \_sync\_faculty\_assignment\_scope.

**What this role can edit/change:** Can change stored fields, relationships, constraints, cascades, and lifecycle behavior. Coordinate schemas, migrations, services, seed data, frontend forms, and tests.

**Change caution:** Data-loss and compatibility risk is high; backup and migration rehearsal are mandatory.

**074. backend/app/models/course.py**

**Purpose:** Defines SQLAlchemy persistence entities and relationships for the course domain.

**Feature contribution:** Adds academic truth, people records, curriculum behavior, or institution hierarchy. Code evidence: tables: courses; classes: Course.

**What this role can edit/change:** Can change stored fields, relationships, constraints, cascades, and lifecycle behavior. Coordinate schemas, migrations, services, seed data, frontend forms, and tests.

**Change caution:** Data-loss and compatibility risk is high; backup and migration rehearsal are mandatory.

**075. backend/app/models/department.py**

**Purpose:** Defines SQLAlchemy persistence entities and relationships for the department domain.

**Feature contribution:** Adds academic truth, people records, curriculum behavior, or institution hierarchy. Code evidence: tables: departments; classes: Department.

**What this role can edit/change:** Can change stored fields, relationships, constraints, cascades, and lifecycle behavior. Coordinate schemas, migrations, services, seed data, frontend forms, and tests.

**Change caution:** Data-loss and compatibility risk is high; backup and migration rehearsal are mandatory.

**080. backend/app/models/student.py**

**Purpose:** Defines SQLAlchemy persistence entities and relationships for the student domain.

**Feature contribution:** Adds academic truth, people records, curriculum behavior, or institution hierarchy. Code evidence: tables: students, student\_sequences; classes: EnrollmentStatus, Student, StudentSequence.

**What this role can edit/change:** Can change stored fields, relationships, constraints, cascades, and lifecycle behavior. Coordinate schemas, migrations, services, seed data, frontend forms, and tests.

**Change caution:** Data-loss and compatibility risk is high; backup and migration rehearsal are mandatory.

**081. backend/app/models/subject.py**

**Purpose:** Defines SQLAlchemy persistence entities and relationships for the subject domain.

**Feature contribution:** Adds academic truth, people records, curriculum behavior, or institution hierarchy. Code evidence: tables: subjects, subject\_teachers; classes: Subject, SubjectTeacher.

**What this role can edit/change:** Can change stored fields, relationships, constraints, cascades, and lifecycle behavior. Coordinate schemas, migrations, services, seed data, frontend forms, and tests.

**Change caution:** Data-loss and compatibility risk is high; backup and migration rehearsal are mandatory.

**082. backend/app/models/teacher.py**

**Purpose:** Defines SQLAlchemy persistence entities and relationships for the teacher domain.

**Feature contribution:** Adds academic truth, people records, curriculum behavior, or institution hierarchy. Code evidence: tables: teachers; classes: Teacher.

**What this role can edit/change:** Can change stored fields, relationships, constraints, cascades, and lifecycle behavior. Coordinate schemas, migrations, services, seed data, frontend forms, and tests.

**Change caution:** Data-loss and compatibility risk is high; backup and migration rehearsal are mandatory.

**090. backend/app/routers/academic.py**

**Purpose:** Publishes FastAPI HTTP endpoints for the academic domain and connects permissions to services/data access.

**Feature contribution:** Adds academic truth, people records, curriculum behavior, or institution hierarchy. Code evidence: top-level functions: `service`, `list_subject_types`, `list_branches`, `create_branch`, `update_branch`, `delete_branch`, `list_curricula`, `create_curriculum`, `update_curriculum`, `delete_curriculum`, `list_curriculum_versions`, `create_curriculum_version`, `update_curriculum_version`, `delete_curriculum_version`; 36 HTTP operations.

**What this role can edit/change:** Can add/change endpoints, filters, status codes, permissions, serialization, and orchestration. Preserve backend authorization and update clients/tests.

**Change caution:** Review downstream imports and the role boundary before merging.

**099. backend/app/schemas/academic.py**

**Purpose:** Defines validated request and response contracts for the academic domain.

**Feature contribution:** Adds academic truth, people records, curriculum behavior, or institution hierarchy. Code evidence: classes: `BranchCreate`, `BranchUpdate`, `CurriculumCreate`, `CurriculumUpdate`, `CurriculumVersionCreate`, `CurriculumVersionUpdate`, `SemesterCreate`, `SemesterUpdate`, `SectionCreate`, `SectionUpdate`, `ElectiveGroupCreate`, `ElectiveGroupUpdate`, `CurriculumSubjectCreate`, `CurriculumSubjectUpdate`.

**What this role can edit/change:** Can change accepted payloads, validation, defaults, and returned fields. Treat response changes as API compatibility changes.

**Change caution:** Review downstream imports and the role boundary before merging.

**104. backend/app/schemas/course.py**

**Purpose:** Defines validated request and response contracts for the course domain.

**Feature contribution:** Adds academic truth, people records, curriculum behavior, or institution hierarchy. Code evidence: classes: `CourseCreate`, `CourseUpdate`, `CourseResponse`.

**What this role can edit/change:** Can change accepted payloads, validation, defaults, and returned fields. Treat response changes as API compatibility changes.

**Change caution:** Review downstream imports and the role boundary before merging.

**106. backend/app/schemas/department.py**

**Purpose:** Defines validated request and response contracts for the department domain.

**Feature contribution:** Adds academic truth, people records, curriculum behavior, or institution hierarchy. Code evidence: classes: `DepartmentCreate`, `DepartmentUpdate`, `DepartmentResponse`.

**What this role can edit/change:** Can change accepted payloads, validation, defaults, and returned fields. Treat response changes as API compatibility changes.

**Change caution:** Review downstream imports and the role boundary before merging.

**110. backend/app/schemas/student.py**

**Purpose:** Defines validated request and response contracts for the student domain.

**Feature contribution:** Adds academic truth, people records, curriculum behavior, or institution hierarchy. Code evidence: classes: `StudentCreate`, `StudentUpdate`, `StudentResponse`.

**What this role can edit/change:** Can change accepted payloads, validation, defaults, and returned fields. Treat response changes as API compatibility changes.

**Change caution:** Review downstream imports and the role boundary before merging.

**111. backend/app/schemas/subject.py**

**Purpose:** Defines validated request and response contracts for the subject domain.

**Feature contribution:** Adds academic truth, people records, curriculum behavior, or institution hierarchy. Code evidence: classes: `SubjectCreate`, `SubjectUpdate`, `SubjectResponse`, `SubjectTeacherAssign`.

**What this role can edit/change:** Can change accepted payloads, validation, defaults, and returned fields. Treat response changes as API compatibility changes.

**Change caution:** Review downstream imports and the role boundary before merging.



**112. backend/app/schemas/teacher.py**

**Purpose:** Defines validated request and response contracts for the teacher domain.

**Feature contribution:** Adds academic truth, people records, curriculum behavior, or institution hierarchy. Code evidence: classes: TeacherCreate, TeacherUpdate, TeacherResponse.

**What this role can edit/change:** Can change accepted payloads, validation, defaults, and returned fields. Treat response changes as API compatibility changes.

**Change caution:** Review downstream imports and the role boundary before merging.

**115. backend/app/services/academic\_service.py**

**Purpose:** Implements academic validation, curriculum resolution, student mapping, sequence allocation, and scoped list behavior.

**Feature contribution:** Adds academic truth, people records, curriculum behavior, or institution hierarchy. Code evidence: classes: AcademicStructureService.

**What this role can edit/change:** Can change business rules, transactions, calculations, workflow state, and domain validation. Add regression tests for every policy change.

**Change caution:** Review downstream imports and the role boundary before merging.

**140. backend/tests/test\_academic\_structure.py**

**Purpose:** Regression tests for test academic structure; protects behavior during future changes.

**Feature contribution:** Adds academic truth, people records, curriculum behavior, or institution hierarchy. Code evidence: top-level functions: academic\_session, test\_common\_branch\_and\_elective\_subject\_resolution\_without\_duplicates, test\_curriculum\_version\_is\_selected\_by\_admission\_batch, test\_timetables\_are\_resolved\_independently\_for\_each\_section, test\_database\_rejects\_duplicate\_common\_subject\_allocations, test\_student\_sequence\_uses\_course\_branch\_format\_and\_survives\_deletion.

**What this role can edit/change:** Can extend expected behavior and fixtures. Do not weaken assertions merely to make a regression pass.

**Change caution:** Review downstream imports and the role boundary before merging.



## 9. Role 5 - Learning, Assessment and Timetable Engineer

Own attendance, marks, assignments, submissions, faculty teaching context, timetable versions, scheduling, and academic reports.

| Accountability | Definition                                                                                                        |
|----------------|-------------------------------------------------------------------------------------------------------------------|
| Owns           | Implementation quality, tests/checks, documentation, review routing, and operational readiness for assigned files |
| May change     | The behavior described in each file card, within API/data/security review rules                                   |
| Must protect   | Role boundaries, backward compatibility, accessibility, data integrity, and traceability                          |
| File count     | 21                                                                                                                |

### Role feature responsibilities

- Faculty portfolio, attendance, marks, assignments, submissions, timetable versions/slots, export utilities, and their migrations/tests.
- Keep learning records tied to authorized teaching context and make schedule state transitions reliable.

### Every file owned by this role

#### 018. backend/alembic/versions/0003\_section\_scoped\_timetables.py

**Purpose:** Versioned database migration for 0003 section scoped timetables; changes persisted schema and/or backfills data.  
**Feature contribution:** Adds teaching delivery, learning evidence, assignment/timetable behavior, or reporting. Code evidence: top-level functions: upgrade, downgrade.  
**What this role can edit/change:** Can change schema history, constraints, indexes, cascades, and backfills. Never rewrite an applied production revision; add a new revision and test upgrade/downgrade.  
**Change caution:** Production databases may already depend on the exact revision graph.

#### 026. backend/alembic/versions/cc9cb668516c\_add\_versioning\_support\_to\_timetables.py

**Purpose:** Versioned database migration for cc9cb668516c add versioning support to timetables; changes persisted schema and/or backfills data.  
**Feature contribution:** Adds teaching delivery, learning evidence, assignment/timetable behavior, or reporting. Code evidence: top-level functions: upgrade, downgrade.  
**What this role can edit/change:** Can change schema history, constraints, indexes, cascades, and backfills. Never rewrite an applied production revision; add a new revision and test upgrade/downgrade.  
**Change caution:** Production databases may already depend on the exact revision graph.

#### 028. backend/alembic/versions/f437010c7643\_add\_student\_assignments.py

**Purpose:** Versioned database migration for f437010c7643 add student assignments; changes persisted schema and/or backfills data.  
**Feature contribution:** Adds teaching delivery, learning evidence, assignment/timetable behavior, or reporting. Code evidence: top-level functions: upgrade, downgrade.  
**What this role can edit/change:** Can change schema history, constraints, indexes, cascades, and backfills. Never rewrite an applied production revision; add a new revision and test upgrade/downgrade.  
**Change caution:** Production databases may already depend on the exact revision graph.

#### 039. backend/app/core/portfolio.py

**Purpose:** Implements faculty allow-list scoping for students, subjects, sections, and assignments.  
**Feature contribution:** Adds teaching delivery, learning evidence, assignment/timetable behavior, or reporting. Code evidence: classes: FacultyPortfolio; top-level functions: is\_admin, is\_scoped\_faculty, get\_faculty\_portfolio, require\_portfolio\_student, require\_portfolio\_subject, require\_portfolio\_context, require\_portfolio\_assignment.  
**What this role can edit/change:** Can change the support behavior described above; coordinate with dependent files and run the relevant build/test workflow.  
**Change caution:** Review downstream imports and the role boundary before merging.

**057. backend/app/management/commands/migrate\_timetable\_grid.py**

**Purpose:** Administrative command for migrate timetable grid; used for controlled data repair, migration, seeding, verification, or cleanup.

**Feature contribution:** Adds teaching delivery, learning evidence, assignment/timetable behavior, or reporting. Code evidence: top-level functions: migrate.

**What this role can edit/change:** Can change one-off operational data behavior. Run only against backups/staging first and make destructive intent explicit.

**Change caution:** Review downstream imports and the role boundary before merging.

**058. backend/app/management/commands/migrate\_timetable\_pdf.py**

**Purpose:** Administrative command for migrate timetable pdf; used for controlled data repair, migration, seeding, verification, or cleanup.

**Feature contribution:** Adds teaching delivery, learning evidence, assignment/timetable behavior, or reporting. Code evidence: top-level functions: migrate.

**What this role can edit/change:** Can change one-off operational data behavior. Run only against backups/staging first and make destructive intent explicit.

**Change caution:** Review downstream imports and the role boundary before merging.

**072. backend/app/models/attendance.py**

**Purpose:** Defines SQLAlchemy persistence entities and relationships for the attendance domain.

**Feature contribution:** Adds teaching delivery, learning evidence, assignment/timetable behavior, or reporting. Code evidence: tables: attendances; classes: AttendanceStatus, Attendance.

**What this role can edit/change:** Can change stored fields, relationships, constraints, cascades, and lifecycle behavior. Coordinate schemas, migrations, services, seed data, frontend forms, and tests.

**Change caution:** Data-loss and compatibility risk is high; backup and migration rehearsal are mandatory.

**078. backend/app/models/marks.py**

**Purpose:** Defines SQLAlchemy persistence entities and relationships for the marks domain.

**Feature contribution:** Adds teaching delivery, learning evidence, assignment/timetable behavior, or reporting. Code evidence: tables: markss, marks; classes: ExamType, Marks.

**What this role can edit/change:** Can change stored fields, relationships, constraints, cascades, and lifecycle behavior. Coordinate schemas, migrations, services, seed data, frontend forms, and tests.

**Change caution:** Data-loss and compatibility risk is high; backup and migration rehearsal are mandatory.

**083. backend/app/models/timetable.py**

**Purpose:** Defines SQLAlchemy persistence entities and relationships for the timetable domain.

**Feature contribution:** Adds teaching delivery, learning evidence, assignment/timetable behavior, or reporting. Code evidence: tables: timetable\_documents; classes: TimetableDocument.

**What this role can edit/change:** Can change stored fields, relationships, constraints, cascades, and lifecycle behavior. Coordinate schemas, migrations, services, seed data, frontend forms, and tests.

**Change caution:** Data-loss and compatibility risk is high; backup and migration rehearsal are mandatory.

**084. backend/app/models/timetable\_grid.py**

**Purpose:** Defines SQLAlchemy persistence entities and relationships for the timetable grid domain.

**Feature contribution:** Adds teaching delivery, learning evidence, assignment/timetable behavior, or reporting. Code evidence: tables: timetable\_versions, timetable\_slots; classes: TimetableVersion, TimetableSlot; top-level functions: \_sync\_timetable\_scope.

**What this role can edit/change:** Can change stored fields, relationships, constraints, cascades, and lifecycle behavior. Coordinate schemas, migrations, services, seed data, frontend forms, and tests.

**Change caution:** Data-loss and compatibility risk is high; backup and migration rehearsal are mandatory.

**092. backend/app/routers/assignments.py**

**Purpose:** Publishes FastAPI HTTP endpoints for the assignments domain and connects permissions to services/data access.

**Feature contribution:** Adds teaching delivery, learning evidence, assignment/timetable behavior, or reporting. Code evidence: top-level functions: get\_assignments, create\_assignment, update\_assignment, delete\_assignment, get\_submissions, submit\_assignment, review\_submission; 7 HTTP operations.

**What this role can edit/change:** Can add/change endpoints, filters, status codes, permissions, serialization, and orchestration. Preserve backend authorization and update clients/tests.

**Change caution:** Review downstream imports and the role boundary before merging.

**100. backend/app/schemas/assignment.py**

**Purpose:** Defines validated request and response contracts for the assignment domain.

**Feature contribution:** Adds teaching delivery, learning evidence, assignment/timetable behavior, or reporting. Code evidence: classes: AssignmentBase, AssignmentCreate, AssignmentUpdate, AssignmentResponse, AssignmentSubmissionBase, AssignmentSubmissionCreate, AssignmentSubmissionReview, AssignmentSubmissionResponse.

**What this role can edit/change:** Can change accepted payloads, validation, defaults, and returned fields. Treat response changes as API compatibility changes.

**Change caution:** Review downstream imports and the role boundary before merging.

**101. backend/app/schemas/attendance.py**

**Purpose:** Defines validated request and response contracts for the attendance domain.

**Feature contribution:** Adds teaching delivery, learning evidence, assignment/timetable behavior, or reporting. Code evidence: classes: AttendanceCreate, AttendanceBulkCreate, AttendanceResponse, AttendanceUpdate, AttendanceSummary.

**What this role can edit/change:** Can change accepted payloads, validation, defaults, and returned fields. Treat response changes as API compatibility changes.

**Change caution:** Review downstream imports and the role boundary before merging.

**109. backend/app/schemas/marks.py**

**Purpose:** Defines validated request and response contracts for the marks domain.

**Feature contribution:** Adds teaching delivery, learning evidence, assignment/timetable behavior, or reporting. Code evidence: classes: MarksCreate, MarksUpdate, MarksResponse.

**What this role can edit/change:** Can change accepted payloads, validation, defaults, and returned fields. Treat response changes as API compatibility changes.

**Change caution:** Review downstream imports and the role boundary before merging.

**116. backend/app/services/assignment\_service.py**

**Purpose:** Encapsulates business logic and reusable operations for assignment service.

**Feature contribution:** Adds teaching delivery, learning evidence, assignment/timetable behavior, or reporting. Code evidence: classes: AssignmentService.

**What this role can edit/change:** Can change business rules, transactions, calculations, workflow state, and domain validation. Add regression tests for every policy change.

**Change caution:** Review downstream imports and the role boundary before merging.

**126. backend/app/utils/\_\_init\_\_.py**

**Purpose:** Marks the directory as a Python package and may expose package-level imports.

**Feature contribution:** Adds teaching delivery, learning evidence, assignment/timetable behavior, or reporting.

**What this role can edit/change:** Can change the support behavior described above; coordinate with dependent files and run the relevant build/test workflow.

**Change caution:** Review downstream imports and the role boundary before merging.

**127. backend/app/utils/csv\_export.py**

**Purpose:** Project support file for csv export.

**Feature contribution:** Adds teaching delivery, learning evidence, assignment/timetable behavior, or reporting. Code evidence: top-level functions: export\_to\_csv.

**What this role can edit/change:** Can change the support behavior described above; coordinate with dependent files and run the relevant build/test workflow.

**Change caution:** Review downstream imports and the role boundary before merging.

**128. backend/app/utils/pdf\_export.py**

**Purpose:** Project support file for pdf export.

**Feature contribution:** Adds teaching delivery, learning evidence, assignment/timetable behavior, or reporting. Code evidence: classes: CMSReport; top-level functions: generate\_pdf\_report.

**What this role can edit/change:** Can change the support behavior described above; coordinate with dependent files and run the relevant build/test workflow.

**Change caution:** Review downstream imports and the role boundary before merging.

**129. backend/app/utils/report\_factory.py**

**Purpose:** Project support file for report factory.

**Feature contribution:** Adds teaching delivery, learning evidence, assignment/timetable behavior, or reporting. Code evidence: classes: ReportGenerator, CSVReportGenerator, PDFReportGenerator, ReportFactory.

**What this role can edit/change:** Can change the support behavior described above; coordinate with dependent files and run the relevant build/test workflow.

**Change caution:** Review downstream imports and the role boundary before merging.

**130. backend/app/utils/serializer.py**

**Purpose:** Project support file for serializer.

**Feature contribution:** Adds teaching delivery, learning evidence, assignment/timetable behavior, or reporting. Code evidence: top-level functions: serialize\_sqlalchemy\_obj.

**What this role can edit/change:** Can change the support behavior described above; coordinate with dependent files and run the relevant build/test workflow.

**Change caution:** Review downstream imports and the role boundary before merging.

**142. backend/tests/test\_faculty\_portfolio.py**

**Purpose:** Regression tests for test faculty portfolio; protects behavior during future changes.

**Feature contribution:** Adds teaching delivery, learning evidence, assignment/timetable behavior, or reporting. Code evidence: top-level functions: portfolio\_session, \_portfolio\_graph, test\_faculty\_portfolio\_contains\_only\_assigned\_students\_subjects\_and\_allocations, test\_cross\_teacher\_portfolio\_context\_is\_forbidden, test\_admin\_only\_role\_checker\_rejects\_teacher\_and\_accepts\_admin, test\_list\_services\_apply\_portfolio\_allow\_lists.

**What this role can edit/change:** Can extend expected behavior and fixtures. Do not weaken assertions merely to make a regression pass.

**Change caution:** Review downstream imports and the role boundary before merging.

## 10. Role 6 - Finance and Library Backend Engineer

Own fee/invoice/payment/expense/salary behavior and the library catalogue, circulation, reservations, and fines.

| Accountability | Definition                                                                                                        |
|----------------|-------------------------------------------------------------------------------------------------------------------|
| Owens          | Implementation quality, tests/checks, documentation, review routing, and operational readiness for assigned files |
| May change     | The behavior described in each file card, within API/data/security review rules                                   |
| Must protect   | Role boundaries, backward compatibility, accessibility, data integrity, and traceability                          |
| File count     | 10                                                                                                                |

### Role feature responsibilities

- Fee configuration, invoices, collections, payments, expenses, salaries, library catalogue/circulation, reference entities, reservations, fines, and related migrations.
- Protect monetary/circulation state, auditability, availability counts, and reconciliation.

### Every file owned by this role

#### 022. backend/alembic/versions/444e3998ae8d\_expand\_library\_entities.py

**Purpose:** Versioned database migration for 444e3998ae8d expand library entities; changes persisted schema and/or backfills data.

**Feature contribution:** Adds financial control or library operations and their persisted/API contracts. Code evidence: top-level functions: upgrade, downgrade.

**What this role can edit/change:** Can change schema history, constraints, indexes, cascades, and backfills. Never rewrite an applied production revision; add a new revision and test upgrade/downgrade.

**Change caution:** Production databases may already depend on the exact revision graph.

#### 024. backend/alembic/versions/86ea4d69a162\_revamp\_finance\_models.py

**Purpose:** Versioned database migration for 86ea4d69a162 revamp finance models; changes persisted schema and/or backfills data.

**Feature contribution:** Adds financial control or library operations and their persisted/API contracts. Code evidence: top-level functions: upgrade, downgrade.

**What this role can edit/change:** Can change schema history, constraints, indexes, cascades, and backfills. Never rewrite an applied production revision; add a new revision and test upgrade/downgrade.

**Change caution:** Production databases may already depend on the exact revision graph.

#### 027. backend/alembic/versions/e13d8371e6e6\_upgrade\_library\_models.py

**Purpose:** Versioned database migration for e13d8371e6e6 upgrade library models; changes persisted schema and/or backfills data.

**Feature contribution:** Adds financial control or library operations and their persisted/API contracts. Code evidence: top-level functions: upgrade, downgrade.

**What this role can edit/change:** Can change schema history, constraints, indexes, cascades, and backfills. Never rewrite an applied production revision; add a new revision and test upgrade/downgrade.

**Change caution:** Production databases may already depend on the exact revision graph.

#### 053. backend/app/management/commands/generate\_mock\_fees.py

**Purpose:** Administrative command for generate mock fees; used for controlled data repair, migration, seeding, verification, or cleanup.

**Feature contribution:** Adds financial control or library operations and their persisted/API contracts. Code evidence: top-level functions: generate\_fees.

**What this role can edit/change:** Can change one-off operational data behavior. Run only against backups/staging first and make destructive intent explicit.

**Change caution:** Review downstream imports and the role boundary before merging.

**076. backend/app/models/finance.py**

**Purpose:** Implements fee structures, invoices, payments, expenses, categories, and staff salary persistence.

**Feature contribution:** Adds financial control or library operations and their persisted/API contracts. Code evidence: tables: fee\_structures, student\_fees, payments, expense\_categories, expenses, staff\_salaries; classes: FeeStatus, PaymentMethod, PaymentStatus, FeeStructure, StudentFee, Payment, ExpenseCategory, Expense, StaffSalary.

**What this role can edit/change:** Can change stored fields, relationships, constraints, cascades, and lifecycle behavior. Coordinate schemas, migrations, services, seed data, frontend forms, and tests.

**Change caution:** Data-loss and compatibility risk is high; backup and migration rehearsal are mandatory.

**077. backend/app/models/library.py**

**Purpose:** Implements catalogue reference data, members, books, circulation, reservations, and fines.

**Feature contribution:** Adds financial control or library operations and their persisted/API contracts. Code evidence: tables: library\_authors, library\_publishers, book\_categories, library\_members, library\_books, book\_issues, book\_reservations, library\_fines; classes: BookStatus, BookCondition, IssueStatus, MemberType, ReservationStatus, LibraryAuthor, LibraryPublisher, BookCategory, LibraryMember, LibraryBook, BookIssue, BookReservation, LibraryFine.

**What this role can edit/change:** Can change stored fields, relationships, constraints, cascades, and lifecycle behavior. Coordinate schemas, migrations, services, seed data, frontend forms, and tests.

**Change caution:** Data-loss and compatibility risk is high; backup and migration rehearsal are mandatory.

**096. backend/app/routers/finance.py**

**Purpose:** Publishes FastAPI HTTP endpoints for the finance domain and connects permissions to services/data access.

**Feature contribution:** Adds financial control or library operations and their persisted/API contracts. Code evidence: top-level functions: \_course\_brief, \_user\_brief, \_fee\_structure, \_student\_fee, \_payment, \_expense, \_salary, \_update, \_delete, \_page, get\_finance\_dashboard\_stats, create\_fee\_structure, list\_fee\_structures, update\_fee\_structure; 28 HTTP operations.

**What this role can edit/change:** Can add/change endpoints, filters, status codes, permissions, serialization, and orchestration. Preserve backend authorization and update clients/tests.

**Change caution:** Money-changing paths require idempotency, reconciliation, audit, and strict authorization.

**107. backend/app/schemas/finance.py**

**Purpose:** Defines validated request and response contracts for the finance domain.

**Feature contribution:** Adds financial control or library operations and their persisted/API contracts. Code evidence: classes: ExpenseCategoryBase, ExpenseCategoryCreate, ExpenseCategoryResponse, ExpenseBase, ExpenseCreate, ExpenseResponse, ExpenseUpdate, FeeStructureBase, FeeStructureCreate, FeeStructureResponse, FeeStructureUpdate, PaymentBase, PaymentCreate, PaymentResponse.

**What this role can edit/change:** Can change accepted payloads, validation, defaults, and returned fields. Treat response changes as API compatibility changes.

**Change caution:** Money-changing paths require idempotency, reconciliation, audit, and strict authorization.

**108. backend/app/schemas/library.py**

**Purpose:** Defines validated request and response contracts for the library domain.

**Feature contribution:** Adds financial control or library operations and their persisted/API contracts. Code evidence: classes: CategoryBase, CategoryCreate, CategoryResponse, AuthorBase, AuthorCreate, AuthorResponse, PublisherBase, PublisherCreate, PublisherResponse, MemberBase, MemberCreate, MemberResponse, BookBase, BookCreate.

**What this role can edit/change:** Can change accepted payloads, validation, defaults, and returned fields. Treat response changes as API compatibility changes.

**Change caution:** Review downstream imports and the role boundary before merging.

**122. backend/app/services/fee\_strategies.py**

**Purpose:** Encapsulates business logic and reusable operations for fee strategies.

**Feature contribution:** Adds financial control or library operations and their persisted/API contracts. Code evidence: classes: FeeCalculationStrategy, StandardFeeStrategy, LatePenaltyStrategy, ScholarshipStrategy.

**What this role can edit/change:** Can change business rules, transactions, calculations, workflow state, and domain validation. Add regression tests for every policy change.

**Change caution:** Review downstream imports and the role boundary before merging.

# 11. Role 7 - Database Reliability, DevOps and Quality Engineer

Own database foundations, repositories, seeding and maintenance, migration operations, deployment assets, tests, logs, and technical documentation.

| Accountability | Definition                                                                                                        |
|----------------|-------------------------------------------------------------------------------------------------------------------|
| Owens          | Implementation quality, tests/checks, documentation, review routing, and operational readiness for assigned files |
| May change     | The behavior described in each file card, within API/data/security review rules                                   |
| Must protect   | Role boundaries, backward compatibility, accessibility, data integrity, and traceability                          |
| File count     | 54                                                                                                                |

## Role feature responsibilities

- Database base/session/repositories, Alembic operations, seed/cleanup/repair commands, runtime packaging, Compose, tests, logs, uploads scaffold, and documentation.
- Make changes repeatable, recoverable, deployable, and verifiable across environments.

## Every file owned by this role

### 001. .gitignore

**Purpose:** Defines repository exclusions for dependencies, secrets, build output, caches, uploads, logs, databases, and internal notes.

**Feature contribution:** Adds data reliability, repeatable operations, deployment, verification, or maintainability.

**What this role can edit/change:** Can change the support behavior described above; coordinate with dependent files and run the relevant build/test workflow.

**Change caution:** Review downstream imports and the role boundary before merging.

### 002. ACADEMIC\_ARCHITECTURE.md

**Purpose:** Authoritative explanation of the normalized academic hierarchy, subject scoping, curriculum locking, and migration intent.

**Feature contribution:** Adds data reliability, repeatable operations, deployment, verification, or maintainability.

**What this role can edit/change:** Can change operator/developer guidance and product explanation. Keep commands and claims synchronized with working code.

**Change caution:** Review downstream imports and the role boundary before merging.

### 003. CURRICULUM\_README.md

**Purpose:** Plain-language curriculum guide explaining curricula, regulation versions, semester subjects, electives, and user setup steps.

**Feature contribution:** Adds data reliability, repeatable operations, deployment, verification, or maintainability.

**What this role can edit/change:** Can change operator/developer guidance and product explanation. Keep commands and claims synchronized with working code.

**Change caution:** Review downstream imports and the role boundary before merging.

### 004. DB\_SETUP.md

**Purpose:** Database lifecycle runbook for fresh initialization, tracked upgrades, backup, restore, validation, and troubleshooting.

**Feature contribution:** Adds data reliability, repeatable operations, deployment, verification, or maintainability.

**What this role can edit/change:** Can change operator/developer guidance and product explanation. Keep commands and claims synchronized with working code.

**Change caution:** Review downstream imports and the role boundary before merging.



**005. DEPLOYMENT.md**

**Purpose:** Deployment options and procedures for VPS, Docker Compose, reverse proxy, TLS, and alternative PaaS hosting.  
**Feature contribution:** Adds data reliability, repeatable operations, deployment, verification, or maintainability.  
**What this role can edit/change:** Can change operator/developer guidance and product explanation. Keep commands and claims synchronized with working code.  
**Change caution:** Review downstream imports and the role boundary before merging.

**006. EXECUTION\_PLAN.md**

**Purpose:** Cross-platform execution plan for Docker and native development, validation, and operational commands.  
**Feature contribution:** Adds data reliability, repeatable operations, deployment, verification, or maintainability.  
**What this role can edit/change:** Can change operator/developer guidance and product explanation. Keep commands and claims synchronized with working code.  
**Change caution:** Review downstream imports and the role boundary before merging.

**007. PPT.md**

**Purpose:** Internal presentation brief defining product narrative, design theme, stack rationale, and OOP storytelling.  
**Feature contribution:** Adds data reliability, repeatable operations, deployment, verification, or maintainability.  
**What this role can edit/change:** Can change operator/developer guidance and product explanation. Keep commands and claims synchronized with working code.  
**Change caution:** Review downstream imports and the role boundary before merging.

**008. README.md**

**Purpose:** Primary product and engineering guide covering capabilities, architecture, setup, roles, API groups, database lifecycle, and security.  
**Feature contribution:** Adds data reliability, repeatable operations, deployment, verification, or maintainability.  
**What this role can edit/change:** Can change operator/developer guidance and product explanation. Keep commands and claims synchronized with working code.  
**Change caution:** Review downstream imports and the role boundary before merging.

**009. SIDEBAR\_ROLES\_ANALYSIS.md**

**Purpose:** Internal reference describing sidebar visibility and expected resources for each application role.  
**Feature contribution:** Adds data reliability, repeatable operations, deployment, verification, or maintainability.  
**What this role can edit/change:** Can change operator/developer guidance and product explanation. Keep commands and claims synchronized with working code.  
**Change caution:** Review downstream imports and the role boundary before merging.

**010. backend/.dockerignore**

**Purpose:** Excludes virtual environments, caches, local secrets, logs, tests, and runtime artifacts from backend container builds.  
**Feature contribution:** Adds data reliability, repeatable operations, deployment, verification, or maintainability.  
**What this role can edit/change:** Can change the support behavior described above; coordinate with dependent files and run the relevant build/test workflow.  
**Change caution:** Review downstream imports and the role boundary before merging.

**011. backend/.env.example**

**Purpose:** Documents backend configuration keys and safe placeholder values without containing live secrets.  
**Feature contribution:** Adds data reliability, repeatable operations, deployment, verification, or maintainability.  
**What this role can edit/change:** Can change the support behavior described above; coordinate with dependent files and run the relevant build/test workflow.  
**Change caution:** Template only. Never place real credentials in this file or the PDF.



**012. backend/Dockerfile**

**Purpose:** Builds the Python backend image, installs pinned dependencies, prepares runtime directories, and starts Uvicorn.

**Feature contribution:** Adds data reliability, repeatable operations, deployment, verification, or maintainability.

**What this role can edit/change:** Can change build, runtime, networking, proxying, migration, or quality behavior. Validate both local and production-like environments.

**Change caution:** Review downstream imports and the role boundary before merging.

**013. backend/alembic.ini**

**Purpose:** Configures Alembic revision paths, logging, and default migration behavior.

**Feature contribution:** Adds data reliability, repeatable operations, deployment, verification, or maintainability.

**What this role can edit/change:** Can change build, runtime, networking, proxying, migration, or quality behavior. Validate both local and production-like environments.

**Change caution:** Review downstream imports and the role boundary before merging.

**014. backend/alembic/env.py**

**Purpose:** Configures online and offline Alembic migration execution against application metadata and the configured database.

**Feature contribution:** Adds data reliability, repeatable operations, deployment, verification, or maintainability. Code evidence: top-level functions: `run_migrations_offline`, `run_migrations_online`.

**What this role can edit/change:** Can change the support behavior described above; coordinate with dependent files and run the relevant build/test workflow.

**Change caution:** Review downstream imports and the role boundary before merging.

**015. backend/alembic/script.py.mako**

**Purpose:** Template used when generating new Alembic revision files.

**Feature contribution:** Adds data reliability, repeatable operations, deployment, verification, or maintainability.

**What this role can edit/change:** Can change the support behavior described above; coordinate with dependent files and run the relevant build/test workflow.

**Change caution:** Review downstream imports and the role boundary before merging.

**016. backend/alembic/versions/0001\_initial\_schema.py**

**Purpose:** Versioned database migration for 0001 initial schema; changes persisted schema and/or backfills data.

**Feature contribution:** Adds data reliability, repeatable operations, deployment, verification, or maintainability. Code evidence: top-level functions: `upgrade`, `downgrade`.

**What this role can edit/change:** Can change schema history, constraints, indexes, cascades, and backfills. Never rewrite an applied production revision; add a new revision and test upgrade/downgrade.

**Change caution:** Production databases may already depend on the exact revision graph.

**021. backend/alembic/versions/2c8000b87f5f\_cascading\_deletes.py**

**Purpose:** Versioned database migration for 2c8000b87f5f cascading deletes; changes persisted schema and/or backfills data.

**Feature contribution:** Adds data reliability, repeatable operations, deployment, verification, or maintainability. Code evidence: top-level functions: `upgrade`, `downgrade`.

**What this role can edit/change:** Can change schema history, constraints, indexes, cascades, and backfills. Never rewrite an applied production revision; add a new revision and test upgrade/downgrade.

**Change caution:** Production databases may already depend on the exact revision graph.

**025. backend/alembic/versions/\_\_\_init\_\_\_py**

**Purpose:** Versioned database migration for init ; changes persisted schema and/or backfills data.

**Feature contribution:** Adds data reliability, repeatable operations, deployment, verification, or maintainability.

**What this role can edit/change:** Can change schema history, constraints, indexes, cascades, and backfills. Never rewrite an applied production revision; add a new revision and test upgrade/downgrade.

**Change caution:** Production databases may already depend on the exact revision graph.

**041. backend/app/database/ \_\_init\_\_.py**

**Purpose:** Marks the directory as a Python package and may expose package-level imports.

**Feature contribution:** Adds data reliability, repeatable operations, deployment, verification, or maintainability.

**What this role can edit/change:** Can change the support behavior described above; coordinate with dependent files and run the relevant build/test workflow.

**Change caution:** Data-loss and compatibility risk is high; backup and migration rehearsal are mandatory.

**042. backend/app/database/base.py**

**Purpose:** Project support file for base.

**Feature contribution:** Adds data reliability, repeatable operations, deployment, verification, or maintainability. Code evidence: classes: Base, TimestampMixin, SoftDeleteMixin, BaseModel.

**What this role can edit/change:** Can change the support behavior described above; coordinate with dependent files and run the relevant build/test workflow.

**Change caution:** Data-loss and compatibility risk is high; backup and migration rehearsal are mandatory.

**043. backend/app/database/seed.py**

**Purpose:** Creates system roles/permissions and a comprehensive demonstration dataset across academic, learning, finance, library, and notifications.

**Feature contribution:** Adds data reliability, repeatable operations, deployment, verification, or maintainability. Code evidence: top-level functions: seed\_system\_data, \_assign\_role, \_user, \_odd\_semesters, \_seed\_is\_current, \_teacher\_name, \_student\_name, \_seed\_academic\_hierarchy, \_seed\_teachers, \_seed\_subjects\_and\_sections, \_seed\_students, \_recent\_working\_days, \_seed\_learning\_records, \_seed\_timetables.

**What this role can edit/change:** Can change the support behavior described above; coordinate with dependent files and run the relevant build/test workflow.

**Change caution:** Data-loss and compatibility risk is high; backup and migration rehearsal are mandatory.

**044. backend/app/database/session.py**

**Purpose:** Project support file for session.

**Feature contribution:** Adds data reliability, repeatable operations, deployment, verification, or maintainability. Code evidence: top-level functions: init\_db.

**What this role can edit/change:** Can change the support behavior described above; coordinate with dependent files and run the relevant build/test workflow.

**Change caution:** Data-loss and compatibility risk is high; backup and migration rehearsal are mandatory.

**046. backend/app/management/ \_\_init\_\_.py**

**Purpose:** Marks the directory as a Python package and may expose package-level imports.

**Feature contribution:** Adds data reliability, repeatable operations, deployment, verification, or maintainability.

**What this role can edit/change:** Can change the support behavior described above; coordinate with dependent files and run the relevant build/test workflow.

**Change caution:** Review downstream imports and the role boundary before merging.

**047. backend/app/management/commands/ \_\_init\_\_.py**

**Purpose:** Administrative command for init ; used for controlled data repair, migration, seeding, verification, or cleanup.

**Feature contribution:** Adds data reliability, repeatable operations, deployment, verification, or maintainability.

**What this role can edit/change:** Can change one-off operational data behavior. Run only against backups/staging first and make destructive intent explicit.

**Change caution:** Review downstream imports and the role boundary before merging.

**048. backend/app/management/commands/add\_column.py**

**Purpose:** Administrative command for add column; used for controlled data repair, migration, seeding, verification, or cleanup.

**Feature contribution:** Adds data reliability, repeatable operations, deployment, verification, or maintainability. Code evidence: top-level functions: main.

**What this role can edit/change:** Can change one-off operational data behavior. Run only against backups/staging first and make destructive intent explicit.

**Change caution:** Review downstream imports and the role boundary before merging.

**049. backend/app/management/commands/add\_parents\_columns.py**

**Purpose:** Administrative command for add parents columns; used for controlled data repair, migration, seeding, verification, or cleanup.

**Feature contribution:** Adds data reliability, repeatable operations, deployment, verification, or maintainability. Code evidence: top-level functions: add\_columns.

**What this role can edit/change:** Can change one-off operational data behavior. Run only against backups/staging first and make destructive intent explicit.

**Change caution:** Review downstream imports and the role boundary before merging.

**050. backend/app/management/commands/create\_mysql\_db.py**

**Purpose:** Administrative command for create mysql db; used for controlled data repair, migration, seeding, verification, or cleanup.

**Feature contribution:** Adds data reliability, repeatable operations, deployment, verification, or maintainability. Code evidence: top-level functions: die, create\_database\_and\_user.

**What this role can edit/change:** Can change one-off operational data behavior. Run only against backups/staging first and make destructive intent explicit.

**Change caution:** Review downstream imports and the role boundary before merging.

**051. backend/app/management/commands/fix\_perms.py**

**Purpose:** Administrative command for fix perms; used for controlled data repair, migration, seeding, verification, or cleanup.

**Feature contribution:** Adds data reliability, repeatable operations, deployment, verification, or maintainability. Code evidence: top-level functions: fix\_permissions.

**What this role can edit/change:** Can change one-off operational data behavior. Run only against backups/staging first and make destructive intent explicit.

**Change caution:** Review downstream imports and the role boundary before merging.

**052. backend/app/management/commands/generate\_demo\_data.py**

**Purpose:** Administrative command for generate demo data; used for controlled data repair, migration, seeding, verification, or cleanup.

**Feature contribution:** Adds data reliability, repeatable operations, deployment, verification, or maintainability. Code evidence: top-level functions: main.

**What this role can edit/change:** Can change one-off operational data behavior. Run only against backups/staging first and make destructive intent explicit.

**Change caution:** Review downstream imports and the role boundary before merging.

**054. backend/app/management/commands/list\_tables.py**

**Purpose:** Administrative command for list tables; used for controlled data repair, migration, seeding, verification, or cleanup.

**Feature contribution:** Adds data reliability, repeatable operations, deployment, verification, or maintainability.

**What this role can edit/change:** Can change one-off operational data behavior. Run only against backups/staging first and make destructive intent explicit.

**Change caution:** Review downstream imports and the role boundary before merging.

**059. backend/app/management/commands/remove\_demo\_data.py**

**Purpose:** Administrative command for remove demo data; used for controlled data repair, migration, seeding, verification, or cleanup.

**Feature contribution:** Adds data reliability, repeatable operations, deployment, verification, or maintainability. Code evidence: top-level functions: main.

**What this role can edit/change:** Can change one-off operational data behavior. Run only against backups/staging first and make destructive intent explicit.

**Change caution:** Potentially destructive or data-replacing operation; never run against valuable data without a verified backup.

**061. backend/app/management/commands/reseed\_exact.py**

**Purpose:** Administrative command for reseed exact; used for controlled data repair, migration, seeding, verification, or cleanup.

**Feature contribution:** Adds data reliability, repeatable operations, deployment, verification, or maintainability. Code evidence: top-level functions: main.

**What this role can edit/change:** Can change one-off operational data behavior. Run only against backups/staging first and make destructive intent explicit.

**Change caution:** Potentially destructive or data-replacing operation; never run against valuable data without a verified backup.

**065. backend/app/management/commands/seed\_master.py**

**Purpose:** Administrative command for seed master; used for controlled data repair, migration, seeding, verification, or cleanup.

**Feature contribution:** Adds data reliability, repeatable operations, deployment, verification, or maintainability. Code evidence: top-level functions: `get_or_create_course`, `get_or_create_department`, `get_or_create_subject`, `seed_courses_and_departments`, `seed_teachers`, `seed_students`, `main`.

**What this role can edit/change:** Can change one-off operational data behavior. Run only against backups/staging first and make destructive intent explicit.

**Change caution:** Potentially destructive or data-replacing operation; never run against valuable data without a verified backup.

**066. backend/app/management/commands/verify\_persistence.py**

**Purpose:** Administrative command for verify persistence; used for controlled data repair, migration, seeding, verification, or cleanup.

**Feature contribution:** Adds data reliability, repeatable operations, deployment, verification, or maintainability. Code evidence: top-level functions: `check_connection`, `check_tables`, `check_constraints`, `check_crud_operations`, `check_transaction_safety`, `check_data_integrity`, `main`.

**What this role can edit/change:** Can change one-off operational data behavior. Run only against backups/staging first and make destructive intent explicit.

**Change caution:** Review downstream imports and the role boundary before merging.

**070. backend/app/models/\_\_init\_\_.py**

**Purpose:** Defines SQLAlchemy persistence entities and relationships for the init domain.

**Feature contribution:** Adds data reliability, repeatable operations, deployment, verification, or maintainability.

**What this role can edit/change:** Can change stored fields, relationships, constraints, cascades, and lifecycle behavior. Coordinate schemas, migrations, services, seed data, frontend forms, and tests.

**Change caution:** Data-loss and compatibility risk is high; backup and migration rehearsal are mandatory.

**086. backend/app/repositories/\_\_init\_\_.py**

**Purpose:** Provides data-access abstractions and queries for init .

**Feature contribution:** Adds data reliability, repeatable operations, deployment, verification, or maintainability.

**What this role can edit/change:** Can change query scope, eager loading, filtering, pagination, and persistence behavior. Review performance and authorization scope.

**Change caution:** Review downstream imports and the role boundary before merging.

**087. backend/app/repositories/base.py**

**Purpose:** Provides data-access abstractions and queries for base.

**Feature contribution:** Adds data reliability, repeatable operations, deployment, verification, or maintainability. Code evidence: classes: `BaseRepository`.

**What this role can edit/change:** Can change query scope, eager loading, filtering, pagination, and persistence behavior. Review performance and authorization scope.

**Change caution:** Review downstream imports and the role boundary before merging.

**088. backend/app/repositories/concrete.py**

**Purpose:** Implements domain repositories and reusable database queries for authorization, people, learning, library, audit, timetable, and notifications.

**Feature contribution:** Adds data reliability, repeatable operations, deployment, verification, or maintainability. Code evidence: classes: `UserRepository`, `RoleRepository`, `PermissionRepository`, `StudentRepository`, `TeacherRepository`, `CourseRepository`, `DepartmentRepository`, `SubjectRepository`, `AttendanceRepository`, `MarksRepository`, `LibraryBookRepository`, `BookIssueRepository`, `AuditLogRepository`, `TimetableVersionRepository`.

**What this role can edit/change:** Can change query scope, eager loading, filtering, pagination, and persistence behavior. Review performance and authorization scope.

**Change caution:** Review downstream imports and the role boundary before merging.

**098. backend/app/schemas/\_\_init\_\_.py**

**Purpose:** Defines validated request and response contracts for the init domain.

**Feature contribution:** Adds data reliability, repeatable operations, deployment, verification, or maintainability.

**What this role can edit/change:** Can change accepted payloads, validation, defaults, and returned fields. Treat response changes as API compatibility changes.

**Change caution:** Review downstream imports and the role boundary before merging.

**114. backend/app/services/ \_\_init\_\_.py**

**Purpose:** Encapsulates business logic and reusable operations for init .

**Feature contribution:** Adds data reliability, repeatable operations, deployment, verification, or maintainability.

**What this role can edit/change:** Can change business rules, transactions, calculations, workflow state, and domain validation. Add regression tests for every policy change.

**Change caution:** Review downstream imports and the role boundary before merging.

**119. backend/app/services/crud\_services.py**

**Purpose:** Encapsulates business logic and reusable operations for crud services.

**Feature contribution:** Adds data reliability, repeatable operations, deployment, verification, or maintainability. Code evidence: classes: UserService, StudentService, TeacherService, CourseService, DepartmentService, SubjectService, AttendanceService, MarksService, LibraryService, AuditService, TimetableGridService.

**What this role can edit/change:** Can change business rules, transactions, calculations, workflow state, and domain validation. Add regression tests for every policy change.

**Change caution:** Review downstream imports and the role boundary before merging.

**131. backend/cleanup.py**

**Purpose:** Project support file for cleanup.

**Feature contribution:** Adds data reliability, repeatable operations, deployment, verification, or maintainability. Code evidence: top-level functions: cleanup.

**What this role can edit/change:** Can change one-off operational data behavior. Run only against backups/staging first and make destructive intent explicit.

**Change caution:** Potentially destructive or data-replacing operation; never run against valuable data without a verified backup.

**132. backend/create\_tables.py**

**Purpose:** Project support file for create tables.

**Feature contribution:** Adds data reliability, repeatable operations, deployment, verification, or maintainability.

**What this role can edit/change:** Can change one-off operational data behavior. Run only against backups/staging first and make destructive intent explicit.

**Change caution:** Review downstream imports and the role boundary before merging.

**133. backend/fix\_api\_routes.py**

**Purpose:** Project support file for fix api routes.

**Feature contribution:** Adds data reliability, repeatable operations, deployment, verification, or maintainability.

**What this role can edit/change:** Can change one-off operational data behavior. Run only against backups/staging first and make destructive intent explicit.

**Change caution:** Review downstream imports and the role boundary before merging.

**134. backend/fix\_lazy.py**

**Purpose:** Project support file for fix lazy.

**Feature contribution:** Adds data reliability, repeatable operations, deployment, verification, or maintainability. Code evidence: top-level functions: fix\_file.

**What this role can edit/change:** Can change one-off operational data behavior. Run only against backups/staging first and make destructive intent explicit.

**Change caution:** Review downstream imports and the role boundary before merging.

**135. backend/logs/cms.log.1**

**Purpose:** Historical runtime diagnostic output retained in the repository; it does not implement a user feature.

**Feature contribution:** Adds data reliability, repeatable operations, deployment, verification, or maintainability.

**What this role can edit/change:** May be replaced by fresh diagnostics, but should not contain secrets or personal data. Prefer removal from version control and managed log retention.

**Change caution:** Repository hygiene concern: historical output can leak operational context and creates noisy diffs.

**136. backend/logs/cms.log.5**

**Purpose:** Historical runtime diagnostic output retained in the repository; it does not implement a user feature.

**Feature contribution:** Adds data reliability, repeatable operations, deployment, verification, or maintainability.

**What this role can edit/change:** May be replaced by fresh diagnostics, but should not contain secrets or personal data. Prefer removal from version control and managed log retention.

**Change caution:** Repository hygiene concern: historical output can leak operational context and creates noisy diffs.

**137. backend/requirements.txt**

**Purpose:** Pins the backend framework, database, migration, validation, authentication, email, file, CAPTCHA, PDF, and test dependencies.

**Feature contribution:** Adds data reliability, repeatable operations, deployment, verification, or maintainability.

**What this role can edit/change:** Can change the support behavior described above; coordinate with dependent files and run the relevant build/test workflow.

**Change caution:** Review downstream imports and the role boundary before merging.

**138. backend/run.py**

**Purpose:** Local backend launcher that starts Uvicorn using application settings.

**Feature contribution:** Adds data reliability, repeatable operations, deployment, verification, or maintainability. Code evidence: top-level functions: main.

**What this role can edit/change:** Can change the support behavior described above; coordinate with dependent files and run the relevant build/test workflow.

**Change caution:** Review downstream imports and the role boundary before merging.

**139. backend/seed\_output.txt**

**Purpose:** Project support file for seed output.

**Feature contribution:** Adds data reliability, repeatable operations, deployment, verification, or maintainability.

**What this role can edit/change:** May be replaced by fresh diagnostics, but should not contain secrets or personal data. Prefer removal from version control and managed log retention.

**Change caution:** Repository hygiene concern: historical output can leak operational context and creates noisy diffs.

**141. backend/tests/test\_app.py**

**Purpose:** Regression tests for test app; protects behavior during future changes.

**Feature contribution:** Adds data reliability, repeatable operations, deployment, verification, or maintainability. Code evidence: top-level functions: test\_core\_api\_routes\_are\_registered, test\_cors\_origins\_are\_normalized, test\_admin\_modules\_expose\_complete\_mutation\_routes, test\_academic\_structure\_exposes\_complete\_management\_routes.

**What this role can edit/change:** Can extend expected behavior and fixtures. Do not weaken assertions merely to make a regression pass.

**Change caution:** Review downstream imports and the role boundary before merging.

**143. backend/tests/test\_live\_workflows.py**

**Purpose:** Regression tests for test live workflows; protects behavior during future changes.

**Feature contribution:** Adds data reliability, repeatable operations, deployment, verification, or maintainability. Code evidence: top-level functions: \_login, \_request, test\_all\_major\_live\_workflows.

**What this role can edit/change:** Can extend expected behavior and fixtures. Do not weaken assertions merely to make a regression pass.

**Change caution:** Review downstream imports and the role boundary before merging.

**144. backend/uploads/.gitkeep**

**Purpose:** Keeps the otherwise empty runtime upload directory represented in version control.

**Feature contribution:** Adds data reliability, repeatable operations, deployment, verification, or maintainability.

**What this role can edit/change:** Only represents directory structure; application upload behavior is changed elsewhere.

**Change caution:** Review downstream imports and the role boundary before merging.

#### 145. docker-compose.yml

**Purpose:** Defines the coordinated MySQL, backend, and frontend container stack plus persistent volumes and health dependency.

**Feature contribution:** Adds data reliability, repeatable operations, deployment, verification, or maintainability.

**What this role can edit/change:** Can change build, runtime, networking, proxying, migration, or quality behavior. Validate both local and production-like environments.

**Change caution:** Review downstream imports and the role boundary before merging.

## 12. Complete coverage index

The index below is a compact audit of the same ownership register. The global number matches the detailed file card. Each path appears once and only once.

| No. | File                                                                          | Primary owner                |
|-----|-------------------------------------------------------------------------------|------------------------------|
| 1   | .gitignore                                                                    | Role 7 - DB, DevOps and QA   |
| 2   | ACADEMIC_ARCHITECTURE.md                                                      | Role 7 - DB, DevOps and QA   |
| 3   | CURRICULUM_README.md                                                          | Role 7 - DB, DevOps and QA   |
| 4   | DB_SETUP.md                                                                   | Role 7 - DB, DevOps and QA   |
| 5   | DEPLOYMENT.md                                                                 | Role 7 - DB, DevOps and QA   |
| 6   | EXECUTION_PLAN.md                                                             | Role 7 - DB, DevOps and QA   |
| 7   | PPT.md                                                                        | Role 7 - DB, DevOps and QA   |
| 8   | README.md                                                                     | Role 7 - DB, DevOps and QA   |
| 9   | SIDEBAR_ROLES_ANALYSIS.md                                                     | Role 7 - DB, DevOps and QA   |
| 10  | backend/.dockerignore                                                         | Role 7 - DB, DevOps and QA   |
| 11  | backend/.env.example                                                          | Role 7 - DB, DevOps and QA   |
| 12  | backend/Dockerfile                                                            | Role 7 - DB, DevOps and QA   |
| 13  | backend/alembic.ini                                                           | Role 7 - DB, DevOps and QA   |
| 14  | backend/alembic/env.py                                                        | Role 7 - DB, DevOps and QA   |
| 15  | backend/alembic/script.py.mako                                                | Role 7 - DB, DevOps and QA   |
| 16  | backend/alembic/versions/0001_initial_schema.py                               | Role 7 - DB, DevOps and QA   |
| 17  | backend/alembic/versions/0002_academic_structure.py                           | Role 4 - Academic Domain     |
| 18  | backend/alembic/versions/0003_section_scoped_timetables.py                    | Role 5 - Learning            |
| 19  | backend/alembic/versions/0004_null_safe_academic_scopes.py                    | Role 4 - Academic Domain     |
| 20  | backend/alembic/versions/0005_canonical_hierarchy_and_student_sequences.py    | Role 4 - Academic Domain     |
| 21  | backend/alembic/versions/2c8000b87f5f_cascading_deletes.py                    | Role 7 - DB, DevOps and QA   |
| 22  | backend/alembic/versions/444e3998ae8d_expand_library_entities.py              | Role 6 - Finance and Library |
| 23  | backend/alembic/versions/812be78ed321_add_hod_id_to_branch_manual.py          | Role 4 - Academic Domain     |
| 24  | backend/alembic/versions/86ea4d69a162_revamp_finance_models.py                | Role 6 - Finance and Library |
| 25  | backend/alembic/versions/__init__.py                                          | Role 7 - DB, DevOps and QA   |
| 26  | backend/alembic/versions/cc9cb668516c_add_versioning_support_to_timetables.py | Role 5 - Learning            |
| 27  | backend/alembic/versions/e13d8371e6e6_upgrade_library_models.py               | Role 6 - Finance and Library |
| 28  | backend/alembic/versions/f437010c7643_add_student_assignments.py              | Role 5 - Learning            |
| 29  | backend/app/__init__.py                                                       | Role 3 - API and Security    |
| 30  | backend/app/cli.py                                                            | Role 3 - API and Security    |
| 31  | backend/app/core/__init__.py                                                  | Role 3 - API and Security    |
| 32  | backend/app/core/config.py                                                    | Role 3 - API and Security    |
| 33  | backend/app/core/dependencies.py                                              | Role 3 - API and Security    |
| 34  | backend/app/core/events.py                                                    | Role 3 - API and Security    |
| 35  | backend/app/core/exception_handlers.py                                        | Role 3 - API and Security    |
| 36  | backend/app/core/exceptions.py                                                | Role 3 - API and Security    |
| 37  | backend/app/core/logging_config.py                                            | Role 3 - API and Security    |
| 38  | backend/app/core/permissions.py                                               | Role 3 - API and Security    |



| No. | File                                                              | Primary owner                |
|-----|-------------------------------------------------------------------|------------------------------|
| 39  | backend/app/core/portfolio.py                                     | Role 5 - Learning            |
| 40  | backend/app/core/security.py                                      | Role 3 - API and Security    |
| 41  | backend/app/database/__init__.py                                  | Role 7 - DB, DevOps and QA   |
| 42  | backend/app/database/base.py                                      | Role 7 - DB, DevOps and QA   |
| 43  | backend/app/database/seed.py                                      | Role 7 - DB, DevOps and QA   |
| 44  | backend/app/database/session.py                                   | Role 7 - DB, DevOps and QA   |
| 45  | backend/app/main.py                                               | Role 3 - API and Security    |
| 46  | backend/app/management/__init__.py                                | Role 7 - DB, DevOps and QA   |
| 47  | backend/app/management/commands/__init__.py                       | Role 7 - DB, DevOps and QA   |
| 48  | backend/app/management/commands/add_column.py                     | Role 7 - DB, DevOps and QA   |
| 49  | backend/app/management/commands/add_parents_columns.py            | Role 7 - DB, DevOps and QA   |
| 50  | backend/app/management/commands/create_mysql_db.py                | Role 7 - DB, DevOps and QA   |
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| 52  | backend/app/management/commands/generate_demo_data.py             | Role 7 - DB, DevOps and QA   |
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| 54  | backend/app/management/commands/list_tables.py                    | Role 7 - DB, DevOps and QA   |
| 55  | backend/app/management/commands/migrate_faculty_ids.py            | Role 4 - Academic Domain     |
| 56  | backend/app/management/commands/migrate_student_ids.py            | Role 4 - Academic Domain     |
| 57  | backend/app/management/commands/migrate_timetable_grid.py         | Role 5 - Learning            |
| 58  | backend/app/management/commands/migrate_timetable_pdf.py          | Role 5 - Learning            |
| 59  | backend/app/management/commands/remove_demo_data.py               | Role 7 - DB, DevOps and QA   |
| 60  | backend/app/management/commands/remove_hod_perms.py               | Role 4 - Academic Domain     |
| 61  | backend/app/management/commands/reseed_exact.py                   | Role 7 - DB, DevOps and QA   |
| 62  | backend/app/management/commands/revert_5_real_life_departments.py | Role 4 - Academic Domain     |
| 63  | backend/app/management/commands/seed_5_real_life_departments.py   | Role 4 - Academic Domain     |
| 64  | backend/app/management/commands/seed_all_subjects.py              | Role 4 - Academic Domain     |
| 65  | backend/app/management/commands/seed_master.py                    | Role 7 - DB, DevOps and QA   |
| 66  | backend/app/management/commands/verify_persistence.py             | Role 7 - DB, DevOps and QA   |
| 67  | backend/app/management/commands/verify_subjects.py                | Role 4 - Academic Domain     |
| 68  | backend/app/middleware/__init__.py                                | Role 3 - API and Security    |
| 69  | backend/app/middleware/audit_middleware.py                        | Role 3 - API and Security    |
| 70  | backend/app/models/__init__.py                                    | Role 7 - DB, DevOps and QA   |
| 71  | backend/app/models/academic.py                                    | Role 4 - Academic Domain     |
| 72  | backend/app/models/attendance.py                                  | Role 5 - Learning            |
| 73  | backend/app/models/audit_log.py                                   | Role 3 - API and Security    |
| 74  | backend/app/models/course.py                                      | Role 4 - Academic Domain     |
| 75  | backend/app/models/department.py                                  | Role 4 - Academic Domain     |
| 76  | backend/app/models/finance.py                                     | Role 6 - Finance and Library |
| 77  | backend/app/models/library.py                                     | Role 6 - Finance and Library |
| 78  | backend/app/models/marks.py                                       | Role 5 - Learning            |
| 79  | backend/app/models/notification.py                                | Role 3 - API and Security    |
| 80  | backend/app/models/student.py                                     | Role 4 - Academic Domain     |

| No. | File                                       | Primary owner                |
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| 81  | backend/app/models/subject.py              | Role 4 - Academic Domain     |
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| 89  | backend/app/routers/__init__.py            | Role 3 - API and Security    |
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| 91  | backend/app/routers/api_routes.py          | Role 3 - API and Security    |
| 92  | backend/app/routers/assignments.py         | Role 5 - Learning            |
| 93  | backend/app/routers/auth.py                | Role 3 - API and Security    |
| 94  | backend/app/routers/captcha.py             | Role 3 - API and Security    |
| 95  | backend/app/routers/dashboard.py           | Role 3 - API and Security    |
| 96  | backend/app/routers/finance.py             | Role 6 - Finance and Library |
| 97  | backend/app/routers/notifications.py       | Role 3 - API and Security    |
| 98  | backend/app/schemas/__init__.py            | Role 7 - DB, DevOps and QA   |
| 99  | backend/app/schemas/academic.py            | Role 4 - Academic Domain     |
| 100 | backend/app/schemas/assignment.py          | Role 5 - Learning            |
| 101 | backend/app/schemas/attendance.py          | Role 5 - Learning            |
| 102 | backend/app/schemas/auth.py                | Role 3 - API and Security    |
| 103 | backend/app/schemas/common.py              | Role 3 - API and Security    |
| 104 | backend/app/schemas/course.py              | Role 4 - Academic Domain     |
| 105 | backend/app/schemas/dashboard.py           | Role 3 - API and Security    |
| 106 | backend/app/schemas/department.py          | Role 4 - Academic Domain     |
| 107 | backend/app/schemas/finance.py             | Role 6 - Finance and Library |
| 108 | backend/app/schemas/library.py             | Role 6 - Finance and Library |
| 109 | backend/app/schemas/marks.py               | Role 5 - Learning            |
| 110 | backend/app/schemas/student.py             | Role 4 - Academic Domain     |
| 111 | backend/app/schemas/subject.py             | Role 4 - Academic Domain     |
| 112 | backend/app/schemas/teacher.py             | Role 4 - Academic Domain     |
| 113 | backend/app/schemas/user.py                | Role 3 - API and Security    |
| 114 | backend/app/services/__init__.py           | Role 7 - DB, DevOps and QA   |
| 115 | backend/app/services/academic_service.py   | Role 4 - Academic Domain     |
| 116 | backend/app/services/assignment_service.py | Role 5 - Learning            |
| 117 | backend/app/services/auth_service.py       | Role 3 - API and Security    |
| 118 | backend/app/services/base.py               | Role 3 - API and Security    |
| 119 | backend/app/services/crud_services.py      | Role 7 - DB, DevOps and QA   |
| 120 | backend/app/services/dashboard_service.py  | Role 3 - API and Security    |
| 121 | backend/app/services/email_service.py      | Role 3 - API and Security    |
| 122 | backend/app/services/fee_strategies.py     | Role 6 - Finance and Library |

| No. | File                                              | Primary owner              |
|-----|---------------------------------------------------|----------------------------|
| 123 | backend/app/services/maintenance_service.py       | Role 3 - API and Security  |
| 124 | backend/app/templates/email/student_welcome.html  | Role 3 - API and Security  |
| 125 | backend/app/templates/email/teacher_welcome.html  | Role 3 - API and Security  |
| 126 | backend/app/utlis/__init__.py                     | Role 5 - Learning          |
| 127 | backend/app/utlis/csv_export.py                   | Role 5 - Learning          |
| 128 | backend/app/utlis/pdf_export.py                   | Role 5 - Learning          |
| 129 | backend/app/utlis/report_factory.py               | Role 5 - Learning          |
| 130 | backend/app/utlis/serializer.py                   | Role 5 - Learning          |
| 131 | backend/cleanup.py                                | Role 7 - DB, DevOps and QA |
| 132 | backend/create_tables.py                          | Role 7 - DB, DevOps and QA |
| 133 | backend/fix_api_routes.py                         | Role 7 - DB, DevOps and QA |
| 134 | backend/fix_lazy.py                               | Role 7 - DB, DevOps and QA |
| 135 | backend/logs/cms.log.1                            | Role 7 - DB, DevOps and QA |
| 136 | backend/logs/cms.log.5                            | Role 7 - DB, DevOps and QA |
| 137 | backend/requirements.txt                          | Role 7 - DB, DevOps and QA |
| 138 | backend/run.py                                    | Role 7 - DB, DevOps and QA |
| 139 | backend/seed_output.txt                           | Role 7 - DB, DevOps and QA |
| 140 | backend/tests/test_academic_structure.py          | Role 4 - Academic Domain   |
| 141 | backend/tests/test_app.py                         | Role 7 - DB, DevOps and QA |
| 142 | backend/tests/test_faculty_portfolio.py           | Role 5 - Learning          |
| 143 | backend/tests/test_live_workflows.py              | Role 7 - DB, DevOps and QA |
| 144 | backend/uploads/.gitkeep                          | Role 7 - DB, DevOps and QA |
| 145 | docker-compose.yml                                | Role 7 - DB, DevOps and QA |
| 146 | frontend/.dockerignore                            | Role 1 - FE Platform       |
| 147 | frontend/Dockerfile                               | Role 1 - FE Platform       |
| 148 | frontend/eslint.config.js                         | Role 1 - FE Platform       |
| 149 | frontend/index.html                               | Role 1 - FE Platform       |
| 150 | frontend/nginx.conf                               | Role 1 - FE Platform       |
| 151 | frontend/package-lock.json                        | Role 1 - FE Platform       |
| 152 | frontend/package.json                             | Role 1 - FE Platform       |
| 153 | frontend/public/institute.png                     | Role 1 - FE Platform       |
| 154 | frontend/src/App.jsx                              | Role 1 - FE Platform       |
| 155 | frontend/src/components/layout/AppShell.jsx       | Role 1 - FE Platform       |
| 156 | frontend/src/components/layout/PermissionGate.jsx | Role 1 - FE Platform       |
| 157 | frontend/src/components/ui/Button.jsx             | Role 1 - FE Platform       |
| 158 | frontend/src/components/ui/Card.jsx               | Role 1 - FE Platform       |
| 159 | frontend/src/components/ui/ConfirmDialog.jsx      | Role 1 - FE Platform       |
| 160 | frontend/src/components/ui/PageHeader.jsx         | Role 1 - FE Platform       |
| 161 | frontend/src/components/ui/PeopleDirectory.jsx    | Role 1 - FE Platform       |
| 162 | frontend/src/components/ui/ProfileDetailModal.jsx | Role 1 - FE Platform       |
| 163 | frontend/src/components/ui/RecordFormModal.jsx    | Role 1 - FE Platform       |
| 164 | frontend/src/components/ui/StatCard.jsx           | Role 1 - FE Platform       |

| No. | File                                      | Primary owner         |
|-----|-------------------------------------------|-----------------------|
| 165 | frontend/src/config/resourceForms.js      | Role 2 - FE Workflows |
| 166 | frontend/src/config/resources.jsx         | Role 2 - FE Workflows |
| 167 | frontend/src/config/roleAccess.js         | Role 2 - FE Workflows |
| 168 | frontend/src/lib/api.js                   | Role 1 - FE Platform  |
| 169 | frontend/src/main.jsx                     | Role 1 - FE Platform  |
| 170 | frontend/src/pages/AssignmentsPage.css    | Role 2 - FE Workflows |
| 171 | frontend/src/pages/AssignmentsPage.jsx    | Role 2 - FE Workflows |
| 172 | frontend/src/pages/AttendancePage.css     | Role 2 - FE Workflows |
| 173 | frontend/src/pages/AttendancePage.jsx     | Role 2 - FE Workflows |
| 174 | frontend/src/pages/DashboardPage.jsx      | Role 2 - FE Workflows |
| 175 | frontend/src/pages/FeeCollectionPage.css  | Role 2 - FE Workflows |
| 176 | frontend/src/pages/FeeCollectionPage.jsx  | Role 2 - FE Workflows |
| 177 | frontend/src/pages/HodManagementPage.jsx  | Role 2 - FE Workflows |
| 178 | frontend/src/pages/HodPanelPage.jsx       | Role 2 - FE Workflows |
| 179 | frontend/src/pages/LoginPage.jsx          | Role 2 - FE Workflows |
| 180 | frontend/src/pages/NotificationsPage.jsx  | Role 2 - FE Workflows |
| 181 | frontend/src/pages/ProfilePage.jsx        | Role 2 - FE Workflows |
| 182 | frontend/src/pages/ResourcePage.jsx       | Role 2 - FE Workflows |
| 183 | frontend/src/pages/TimetablePage.jsx      | Role 2 - FE Workflows |
| 184 | frontend/src/state/AuthContext.jsx        | Role 1 - FE Platform  |
| 185 | frontend/src/styles/dashboard-widgets.css | Role 1 - FE Platform  |
| 186 | frontend/src/styles/index.css             | Role 1 - FE Platform  |
| 187 | frontend/src/styles/people-directory.css  | Role 1 - FE Platform  |
| 188 | frontend/vite.config.js                   | Role 1 - FE Platform  |

## 13. Change-request routing examples

| Request                           | Lead files/role                        | Required coordination                                                                     |
|-----------------------------------|----------------------------------------|-------------------------------------------------------------------------------------------|
| Add parent portal                 | Role 2 pages/config; Role 3 auth/API   | Role 4 student relationship; Role 7 migration/tests                                       |
| Change student ID format          | Role 4 student model/service/migration | Role 7 backfill/rollback; Role 2 form/display; Role 3 API contract                        |
| Add payment gateway               | Role 6 finance routes/models           | Role 3 security/audit; Role 2 checkout UX; Role 7 secrets/deployment/reconciliation tests |
| Add timetable collision detection | Role 5 timetable service/model         | Role 4 academic constraints; Role 2 timetable UX; Role 7 performance tests                |
| Add new campus                    | Role 4 domain design                   | Role 3 authorization scope; Role 7 tenant migration/deployment; both frontend roles       |
| Upgrade React/FastAPI/MySQL       | Role 1 or 7 by layer                   | All affected owners run compatibility and regression review                               |

## 14. Definition of done by role

- Role 1: frontend lint/build pass; shared components remain accessible and responsive; session failures are handled.
- Role 2: workflow supports loading, empty, validation, success, and error states; role visibility matches backend policy.

- Role 3: endpoint contract documented; authentication/authorization tested; errors and audit behavior verified.
- Role 4: academic invariants, batch/branch/null scope, and identifiers covered by migration and regression tests.
- Role 5: teacher/student scope and workflow state transitions tested; timetable or assessment data remains consistent.
- Role 6: money/copy state is transactionally correct, reconcilable, authorized, and auditable.
- Role 7: migration rehearsed, backup/restore considered, deployment documentation updated, and automated checks pass.
- All roles: no secret or personal data added to source, logs, fixtures, screenshots, or generated documents.